

BFS Traversal

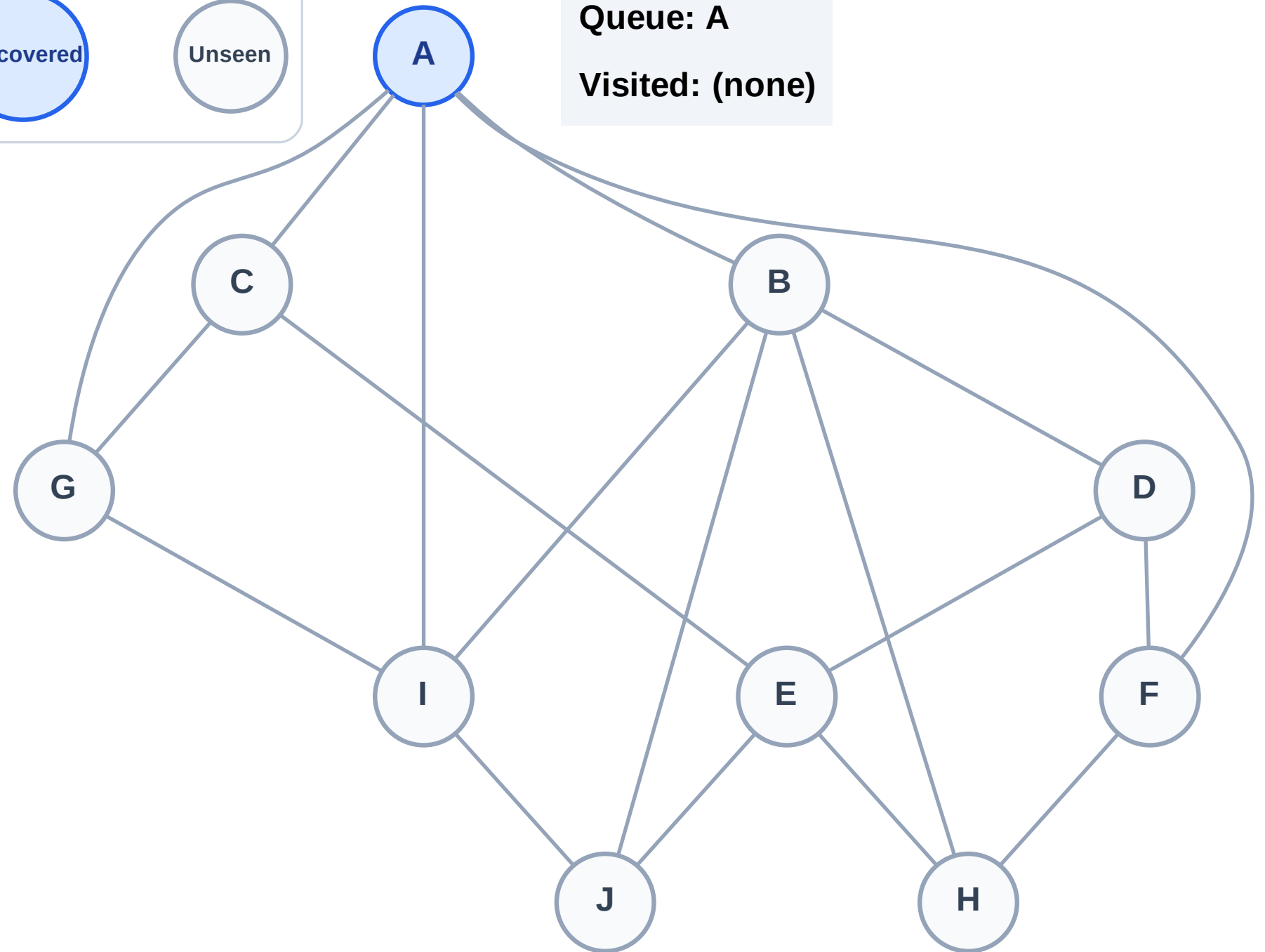
Step 1: Start at A

Legend



Queue: A

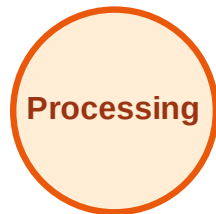
Visited: (none)



BFS Traversal

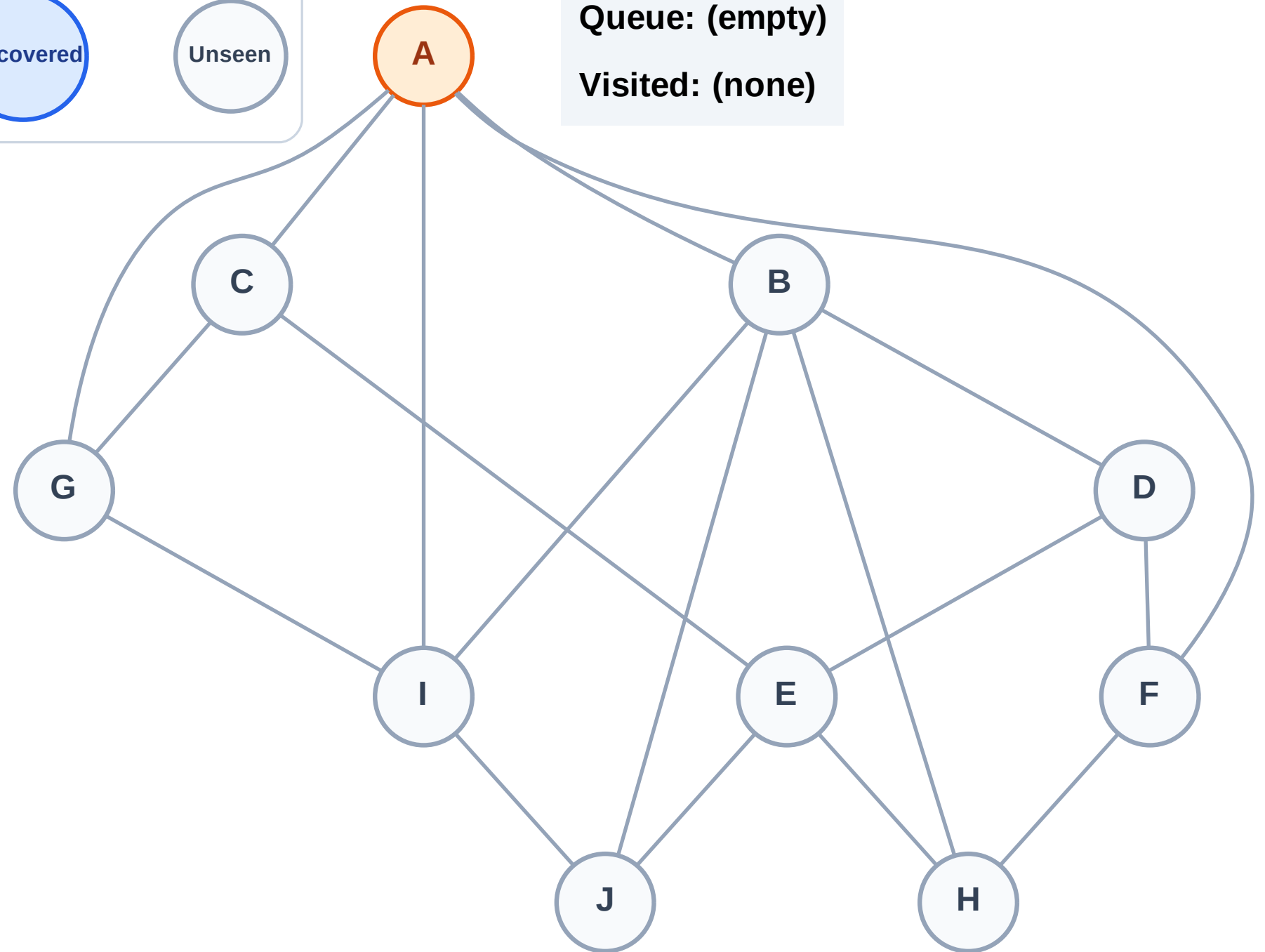
Step 2: Process node A

Legend



Queue: (empty)

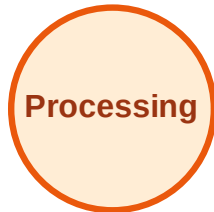
Visited: (none)



BFS Traversal

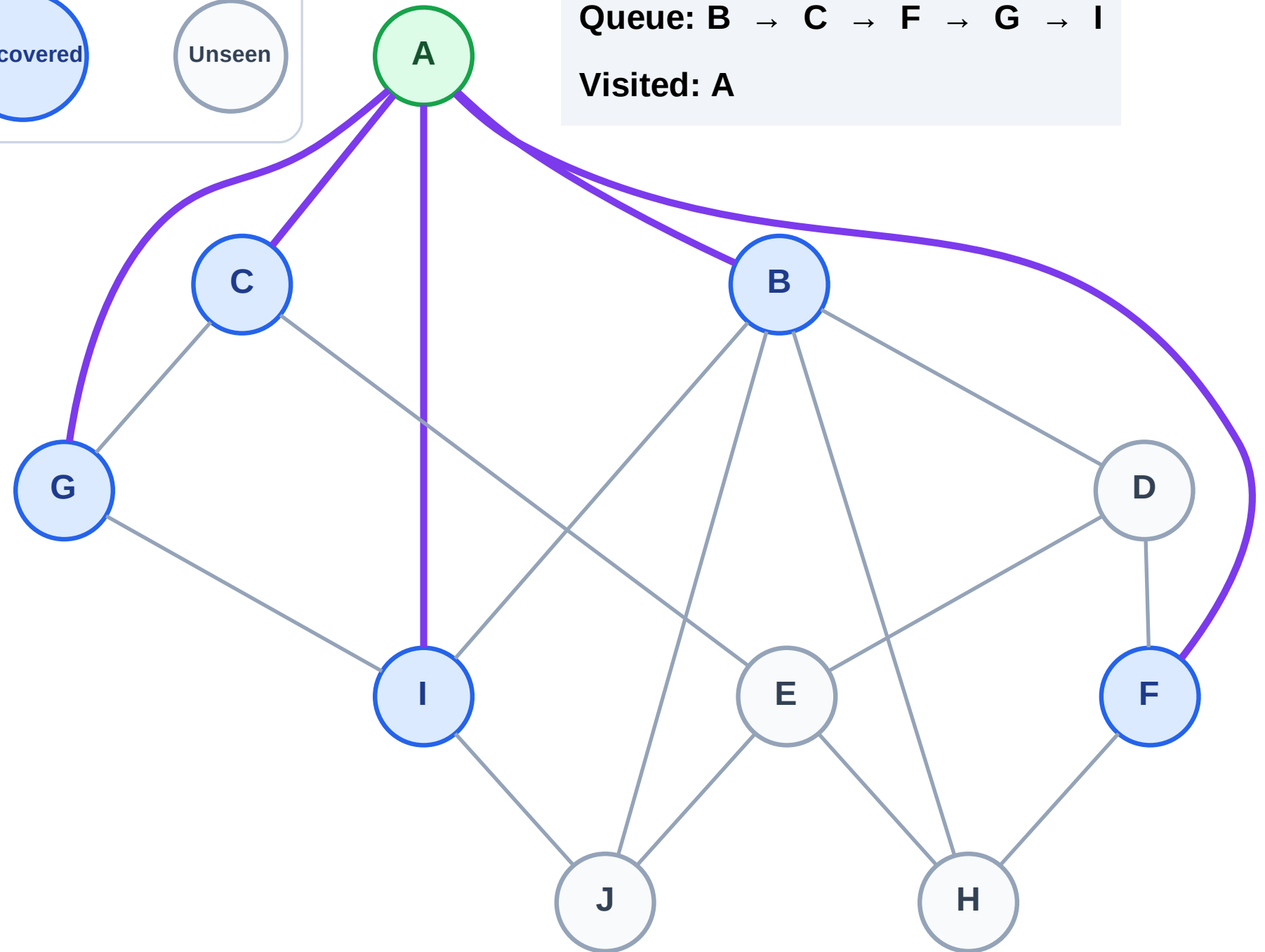
Step 3: Discover B, C, F, G, I from A

Legend



Queue: B → C → F → G → I

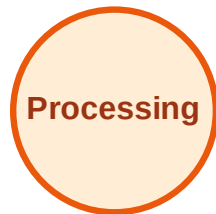
Visited: A



BFS Traversal

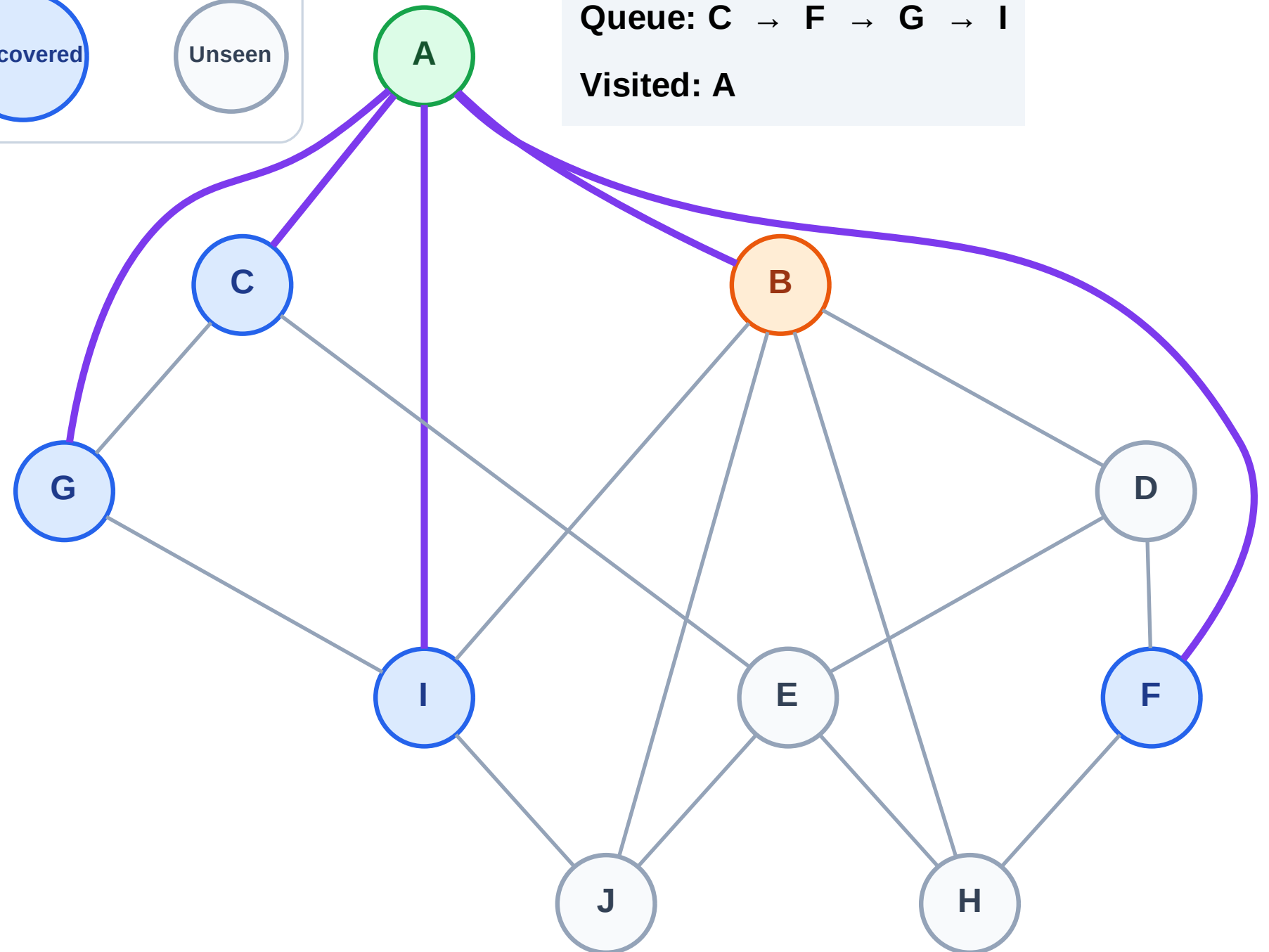
Step 4: Process node B

Legend



Queue: C → F → G → I

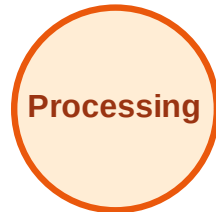
Visited: A



BFS Traversal

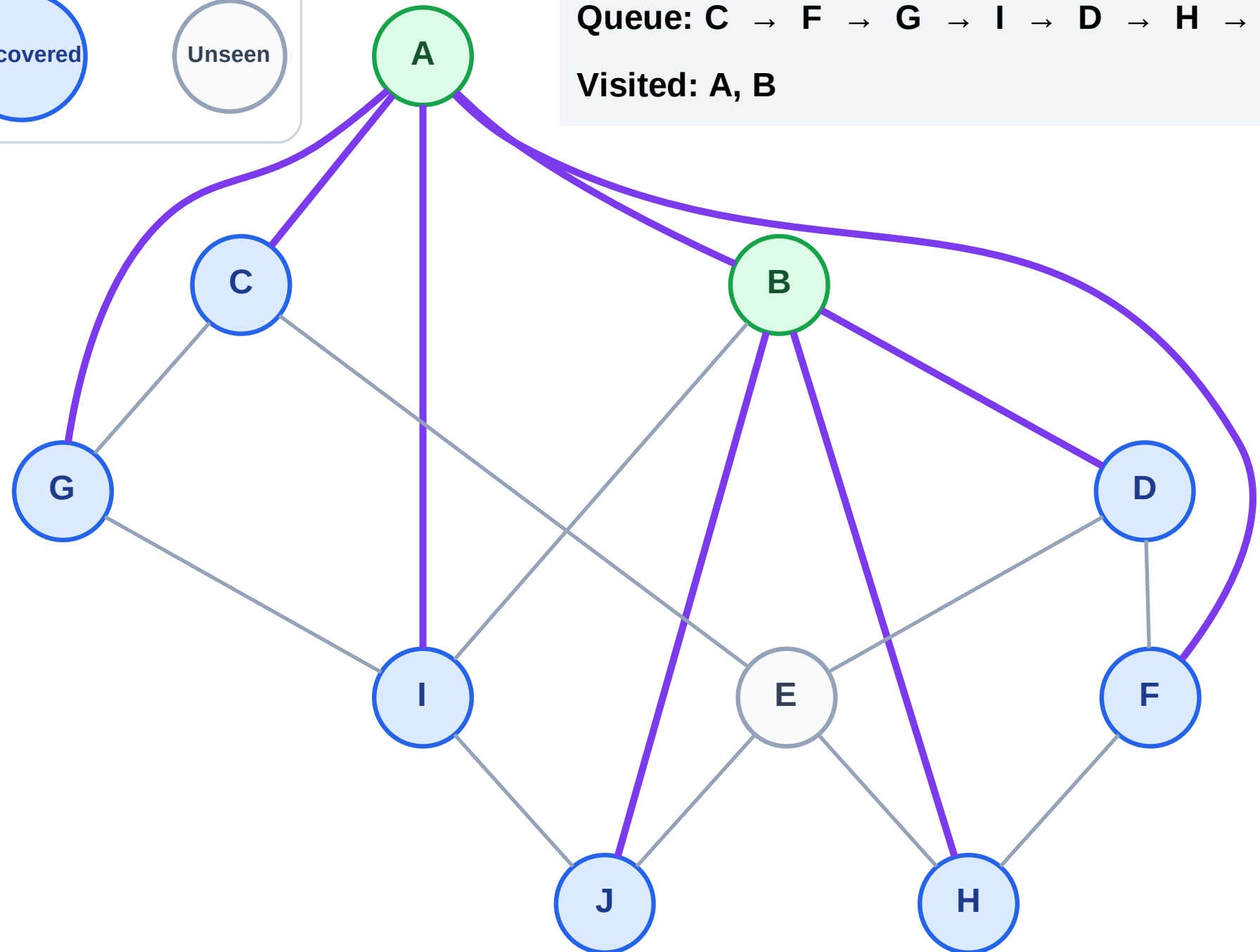
Step 5: Discover D, H, J from B

Legend



Queue: C → F → G → I → D → H → J

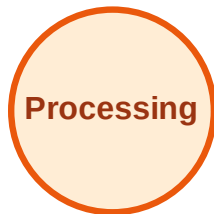
Visited: A, B



BFS Traversal

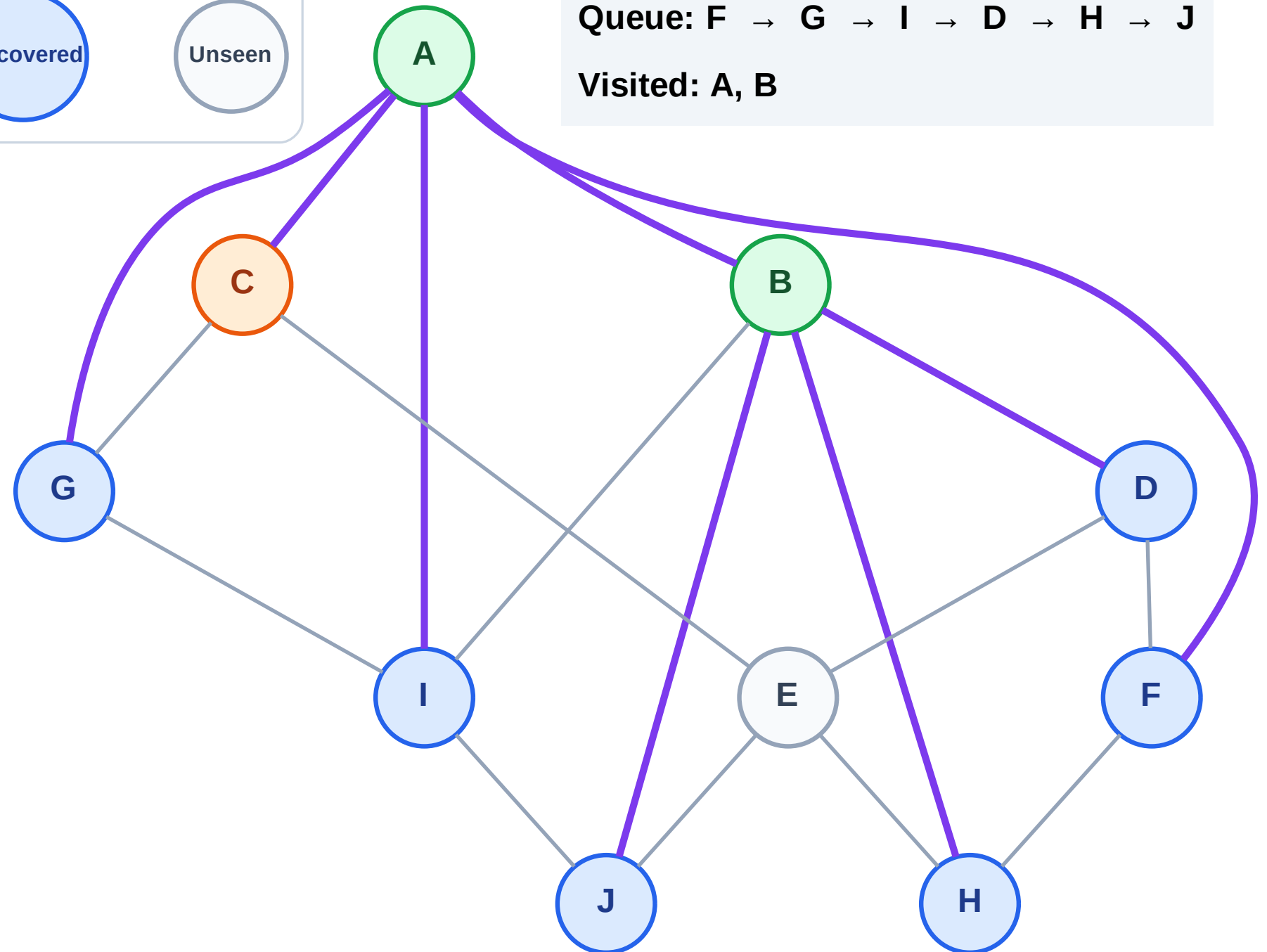
Step 6: Process node C

Legend



Queue: F → G → I → D → H → J

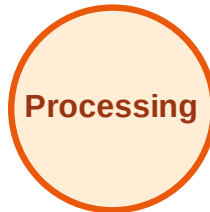
Visited: A, B



BFS Traversal

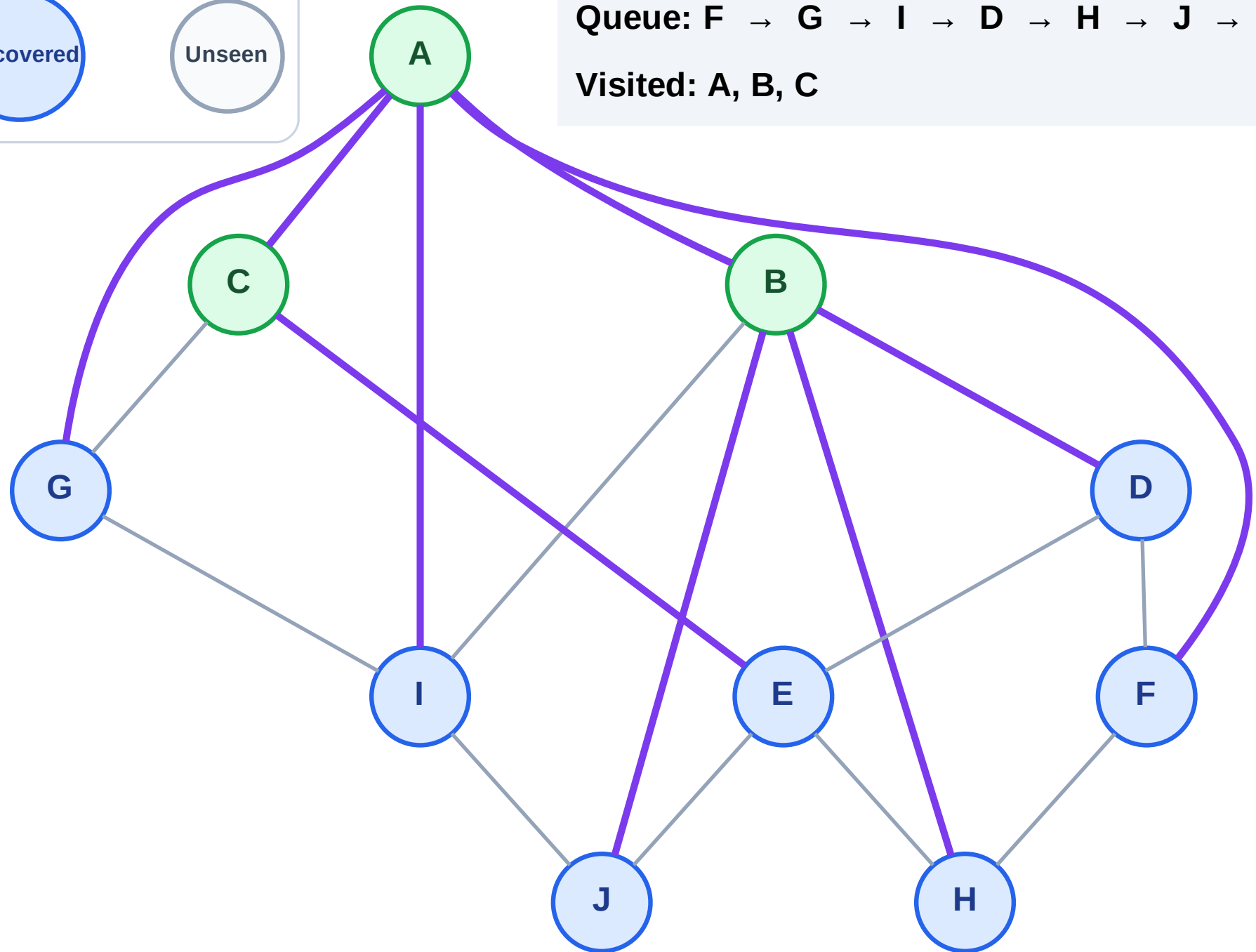
Step 7: Discover E from C

Legend



Queue: F → G → I → D → H → J → E

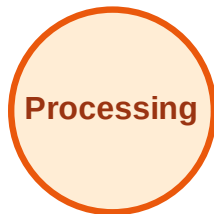
Visited: A, B, C



BFS Traversal

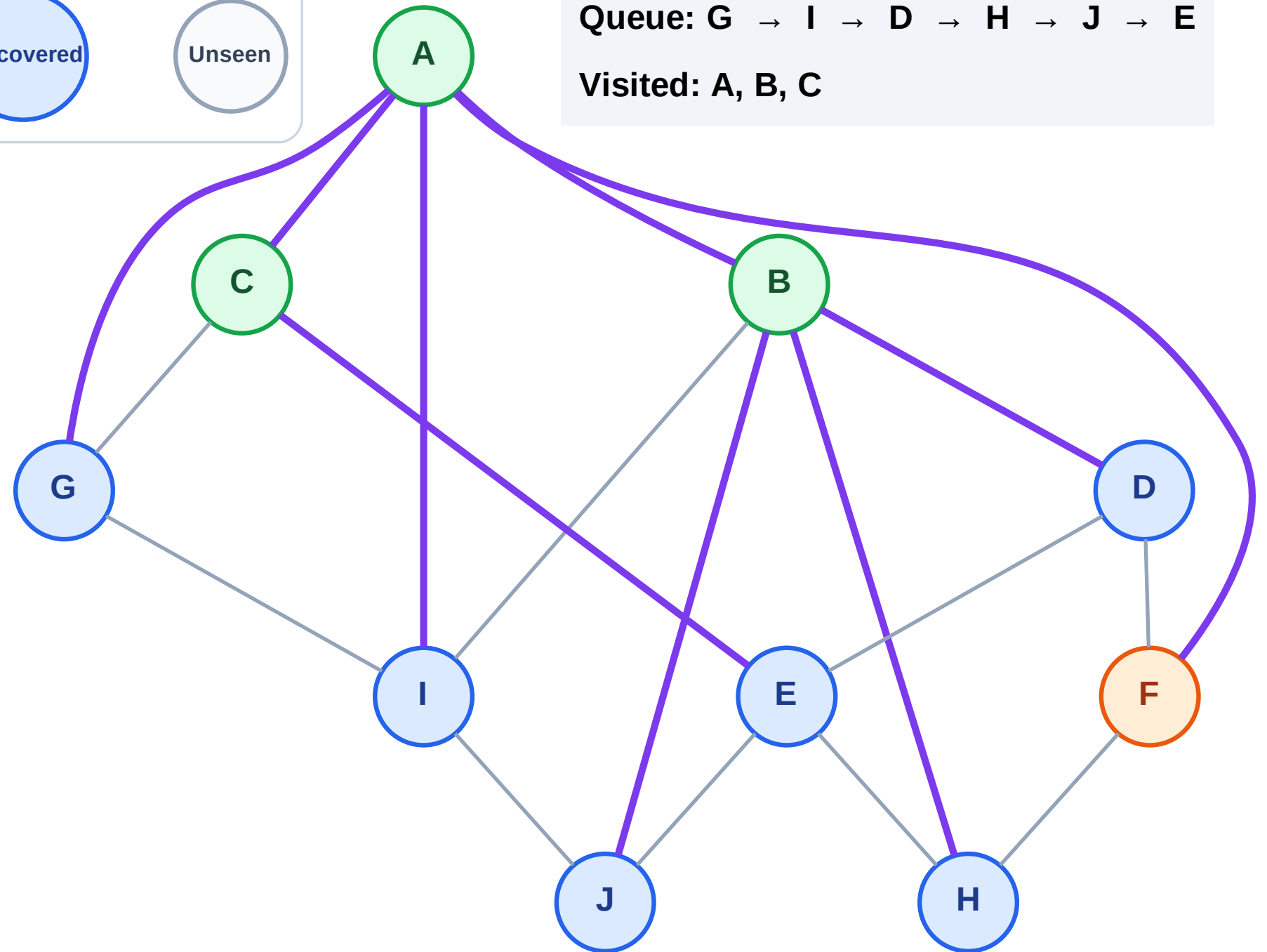
Step 8: Process node F

Legend



Queue: G → I → D → H → J → E

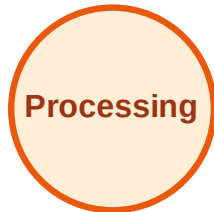
Visited: A, B, C



BFS Traversal

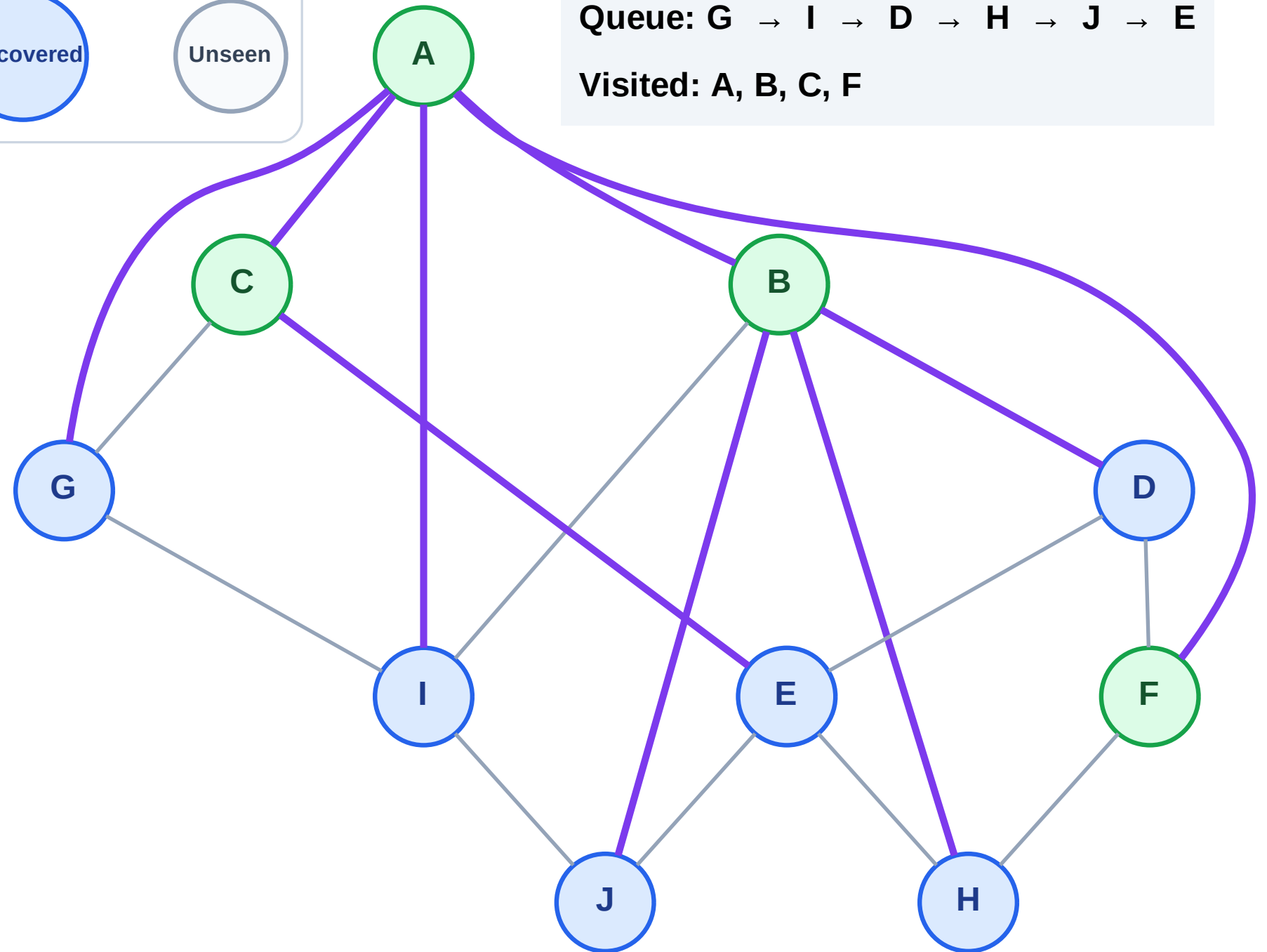
Step 9: No new neighbors from F

Legend



Queue: G → I → D → H → J → E

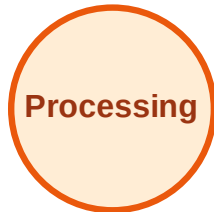
Visited: A, B, C, F



BFS Traversal

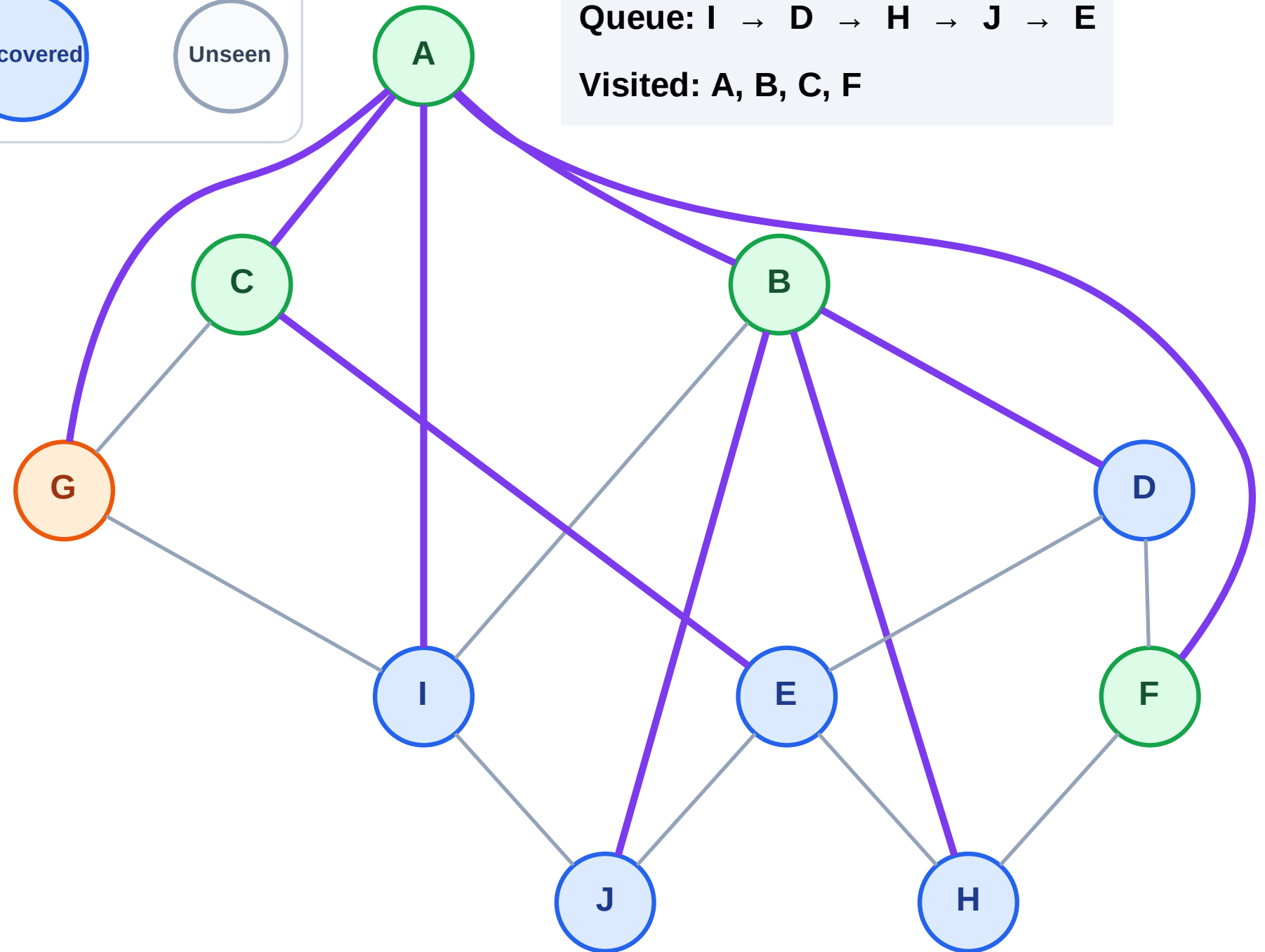
Step 10: Process node G

Legend



Queue: I → D → H → J → E

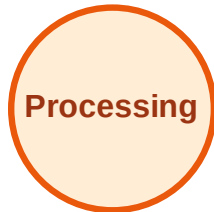
Visited: A, B, C, F



BFS Traversal

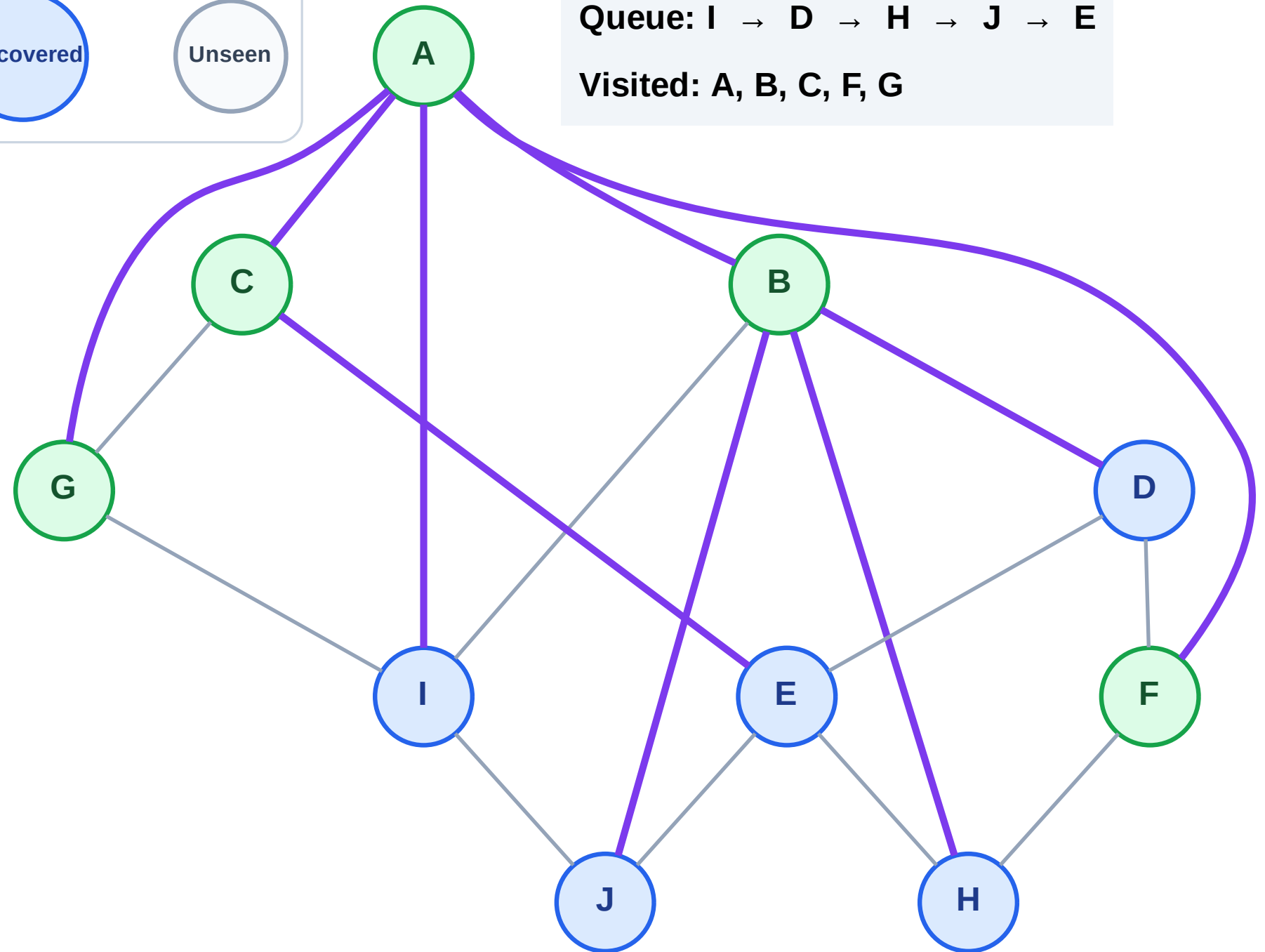
Step 11: No new neighbors from G

Legend



Queue: I → D → H → J → E

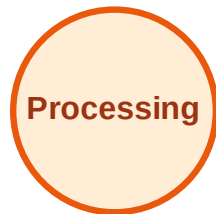
Visited: A, B, C, F, G



BFS Traversal

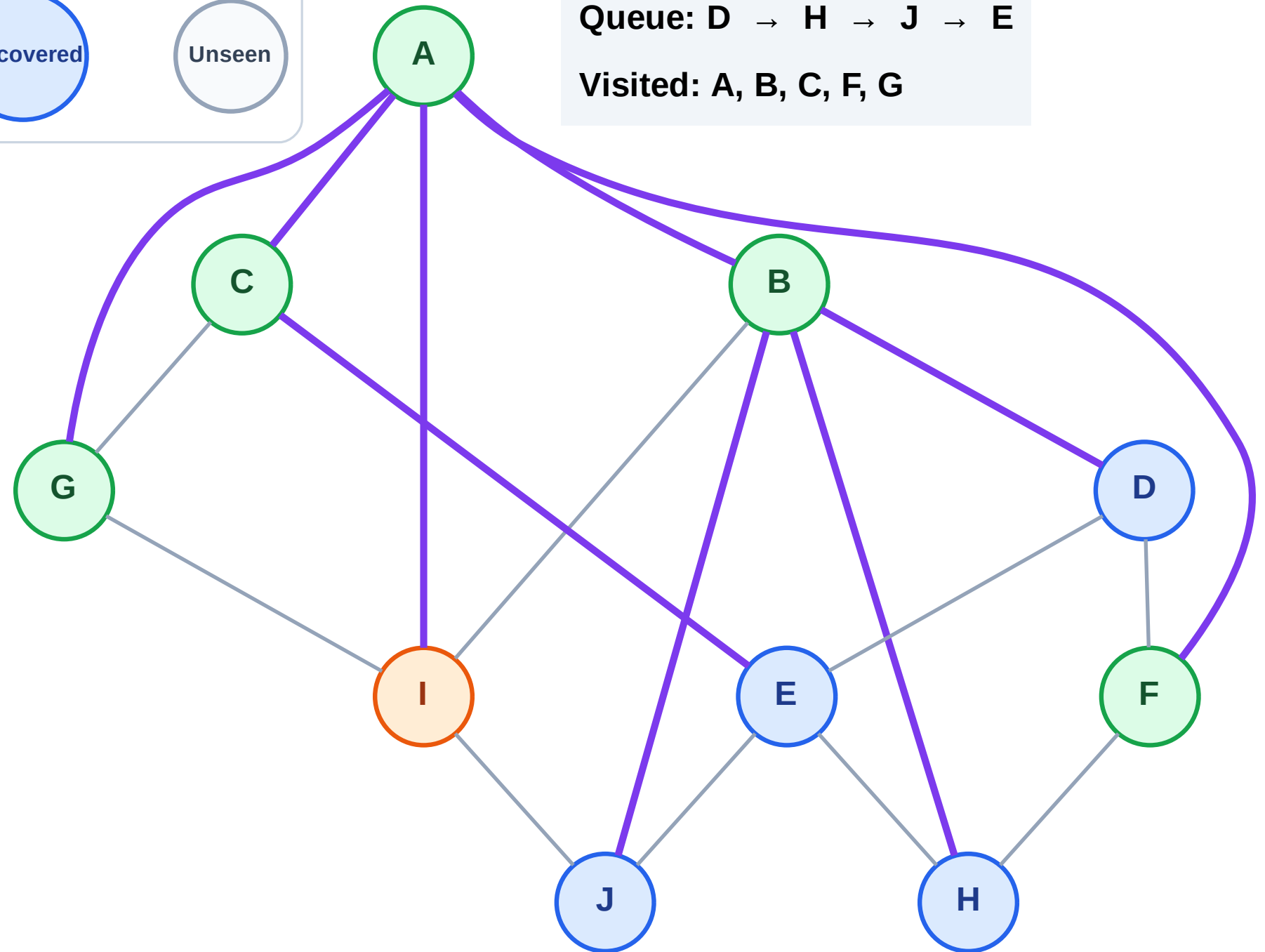
Step 12: Process node I

Legend



Queue: D → H → J → E

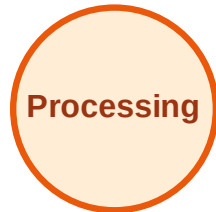
Visited: A, B, C, F, G



BFS Traversal

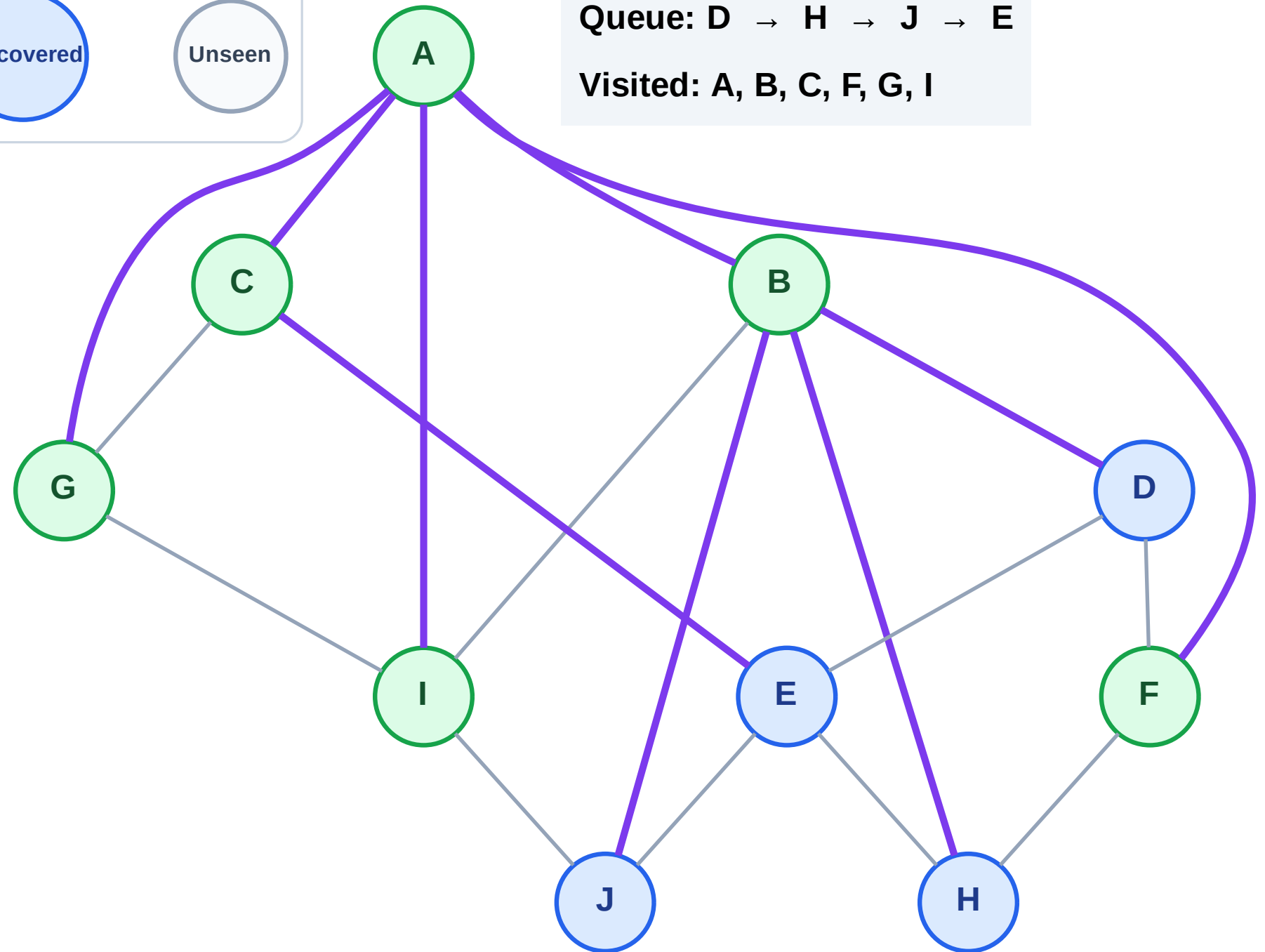
Step 13: No new neighbors from I

Legend



Queue: D → H → J → E

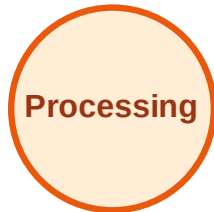
Visited: A, B, C, F, G, I



BFS Traversal

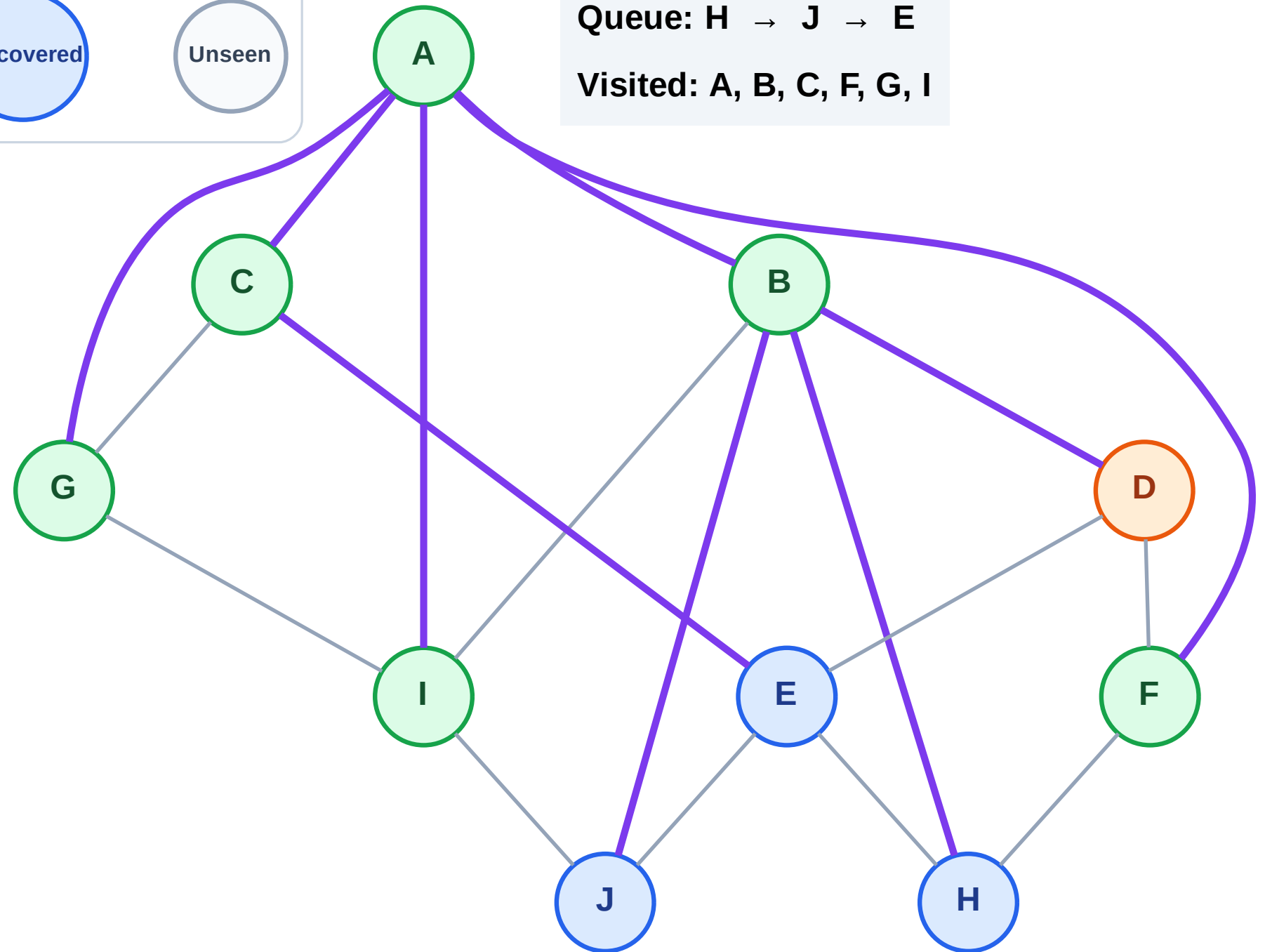
Step 14: Process node D

Legend



Queue: H → J → E

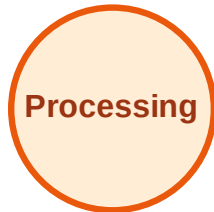
Visited: A, B, C, F, G, I



BFS Traversal

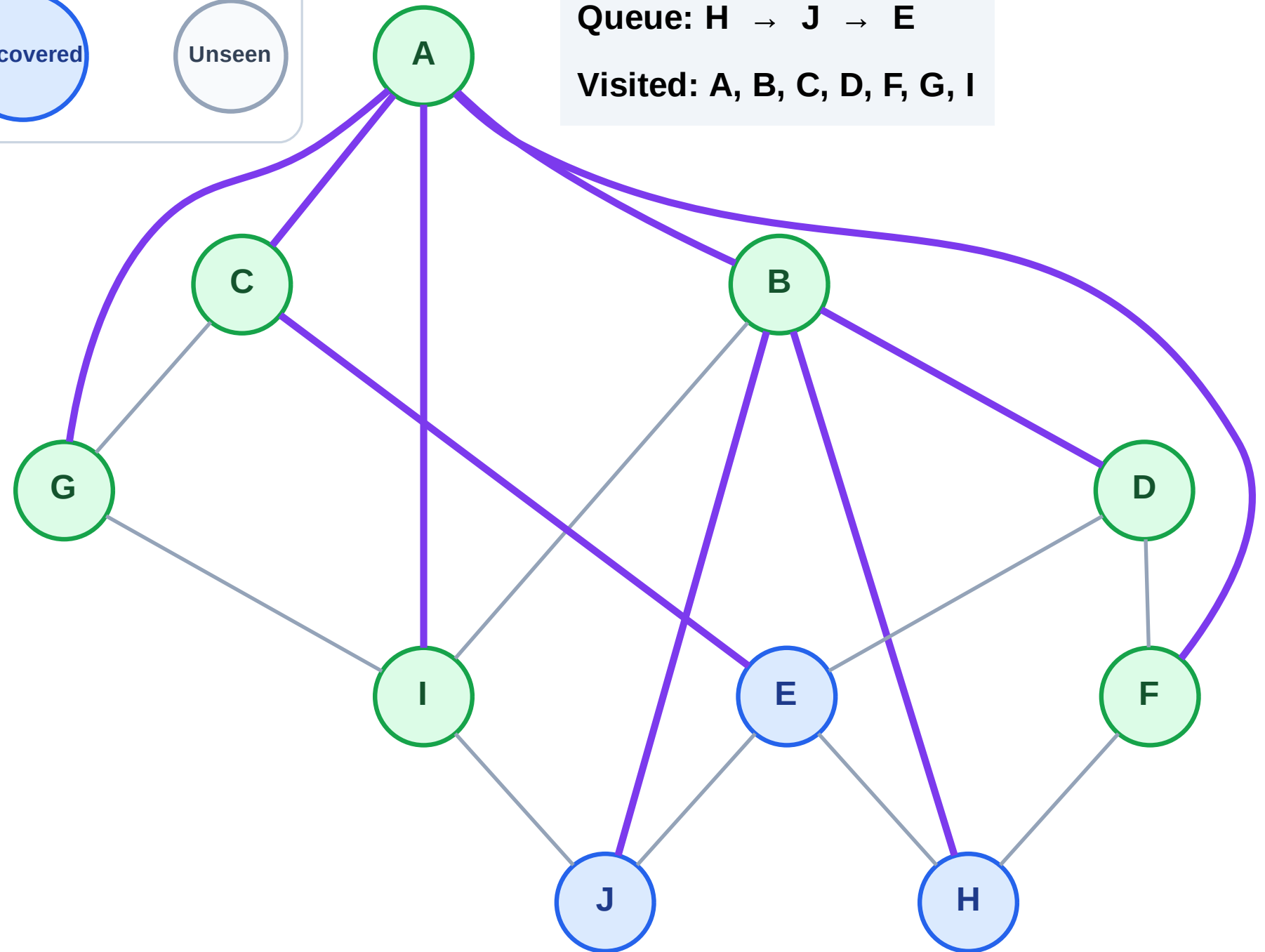
Step 15: No new neighbors from D

Legend



Queue: H → J → E

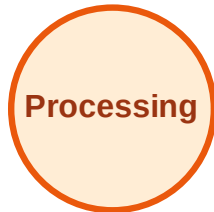
Visited: A, B, C, D, F, G, I



BFS Traversal

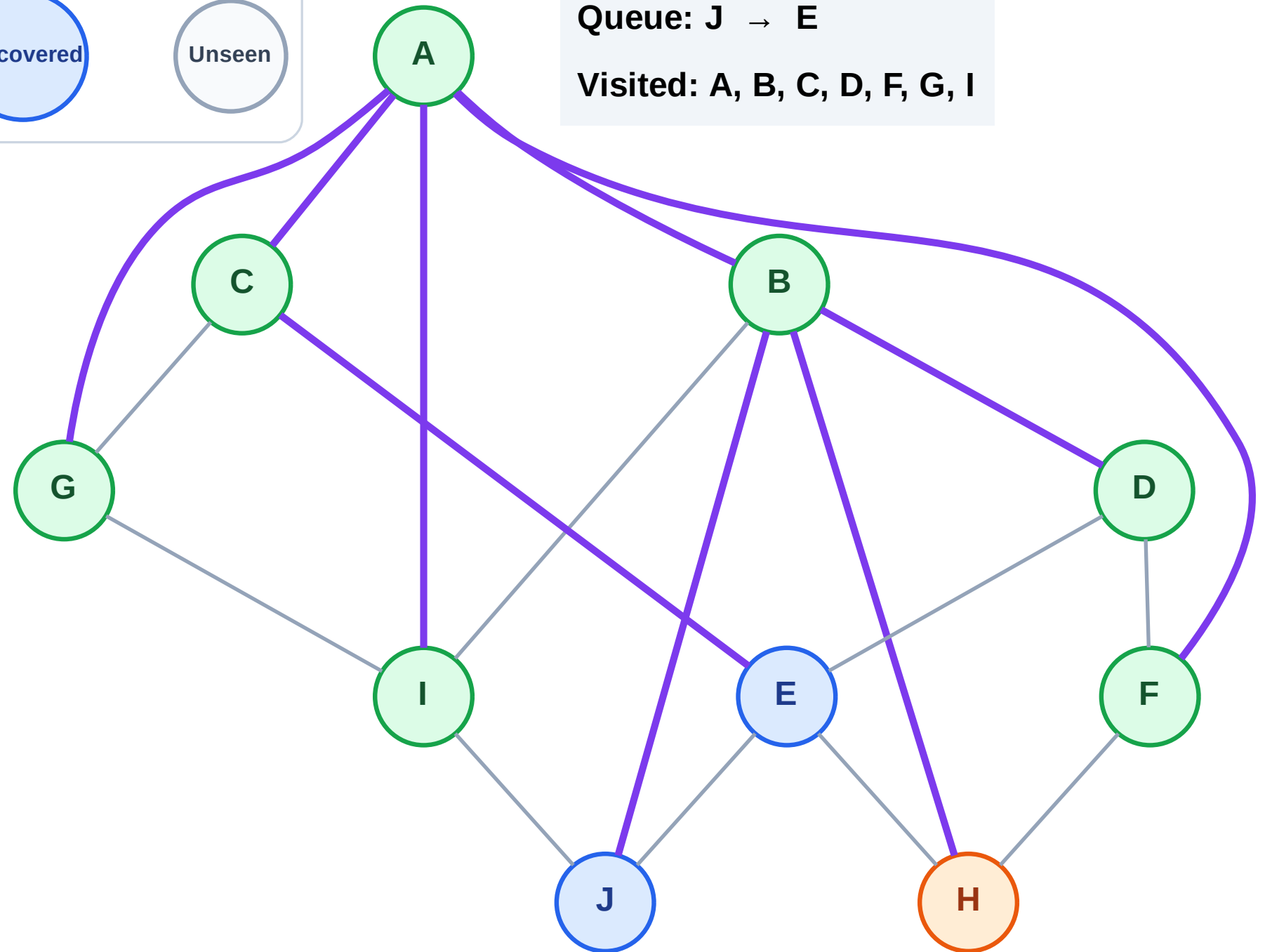
Step 16: Process node H

Legend



Queue: J → E

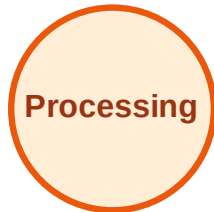
Visited: A, B, C, D, F, G, I



BFS Traversal

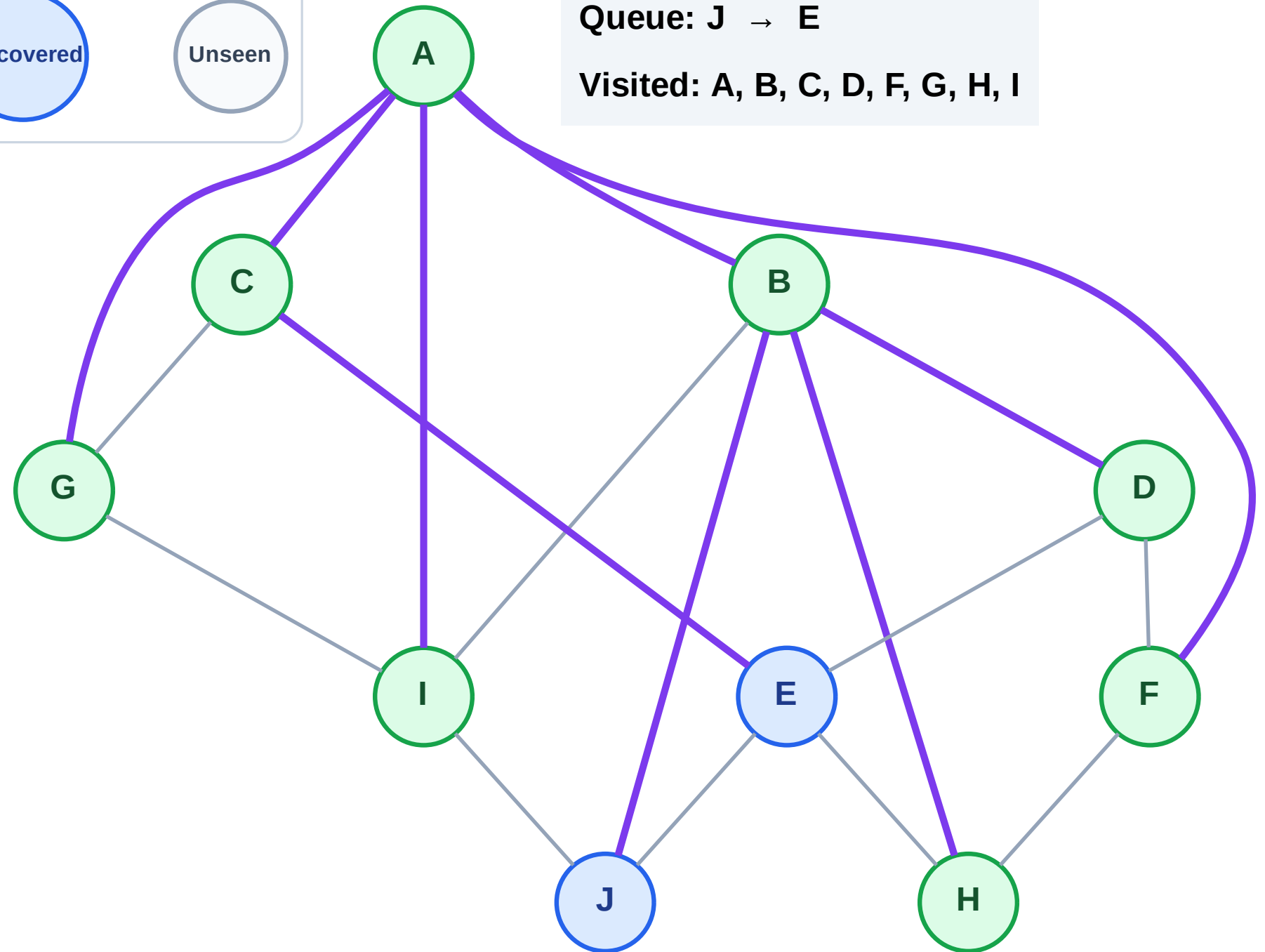
Step 17: No new neighbors from H

Legend



Queue: J → E

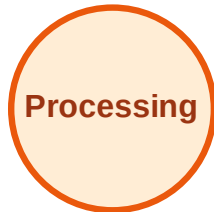
Visited: A, B, C, D, F, G, H, I



BFS Traversal

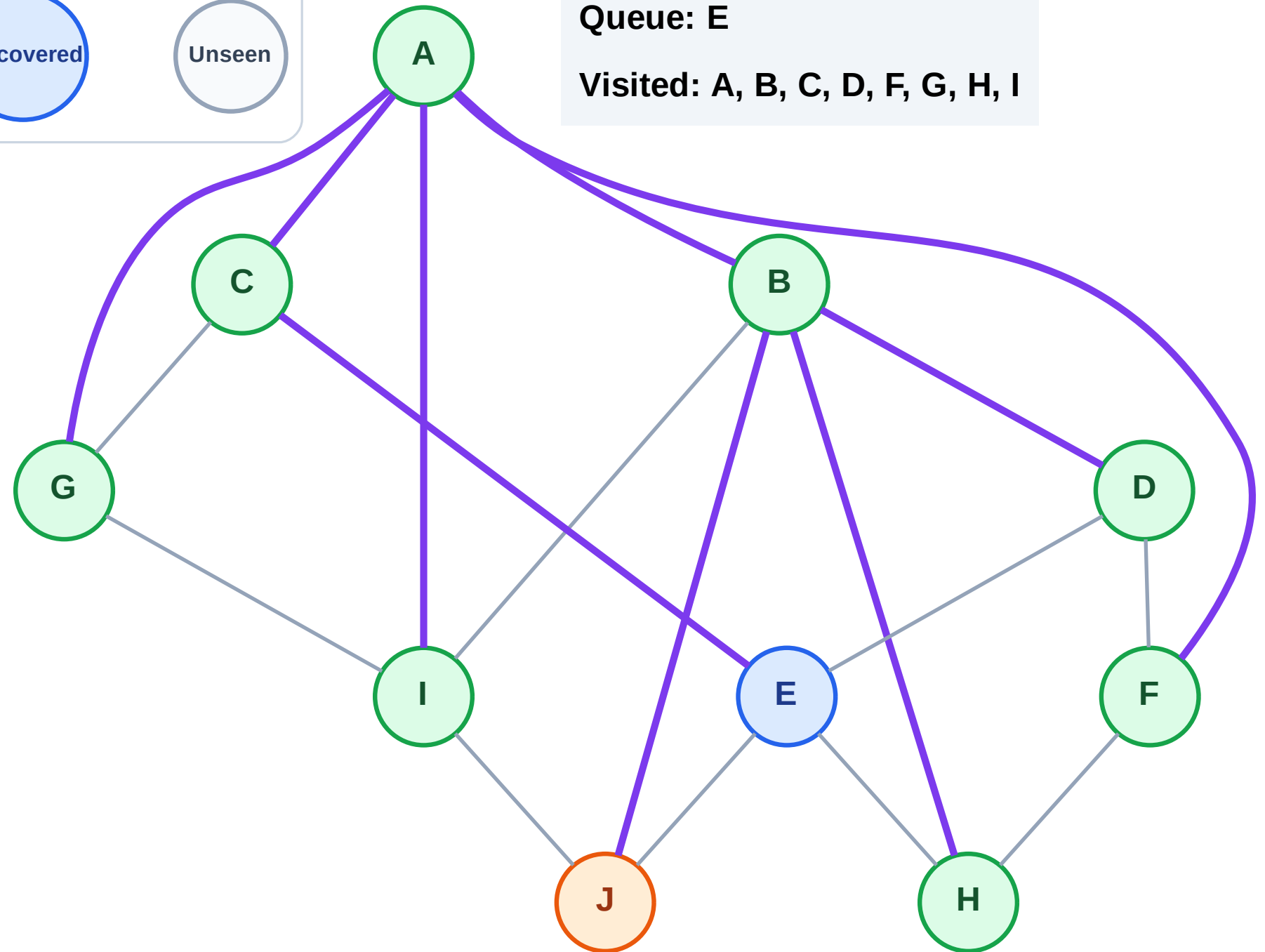
Step 18: Process node J

Legend



Queue: E

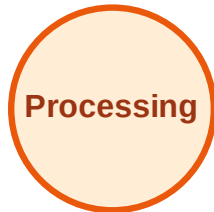
Visited: A, B, C, D, F, G, H, I



BFS Traversal

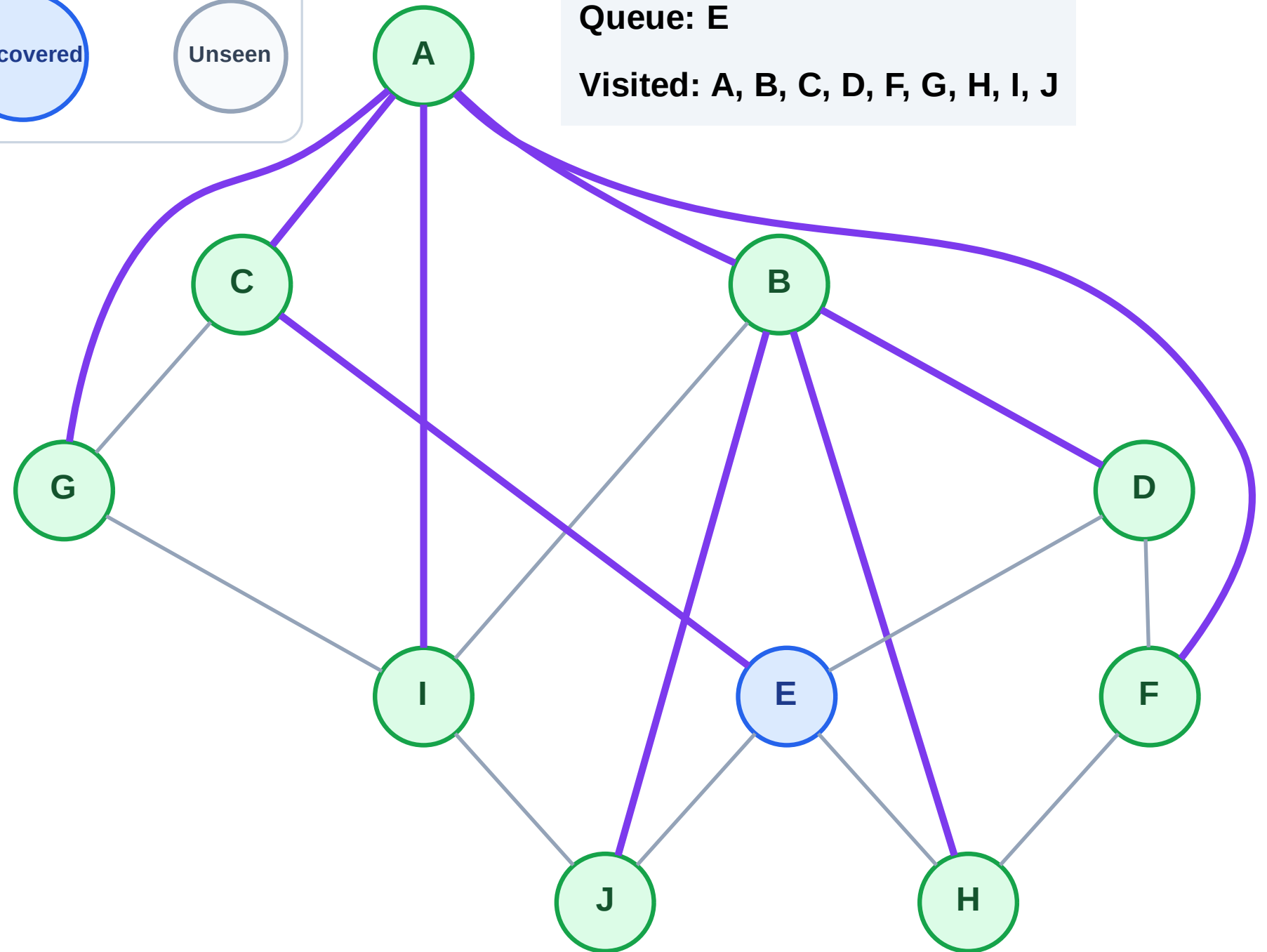
Step 19: No new neighbors from J

Legend



Queue: E

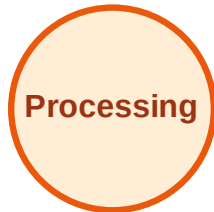
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

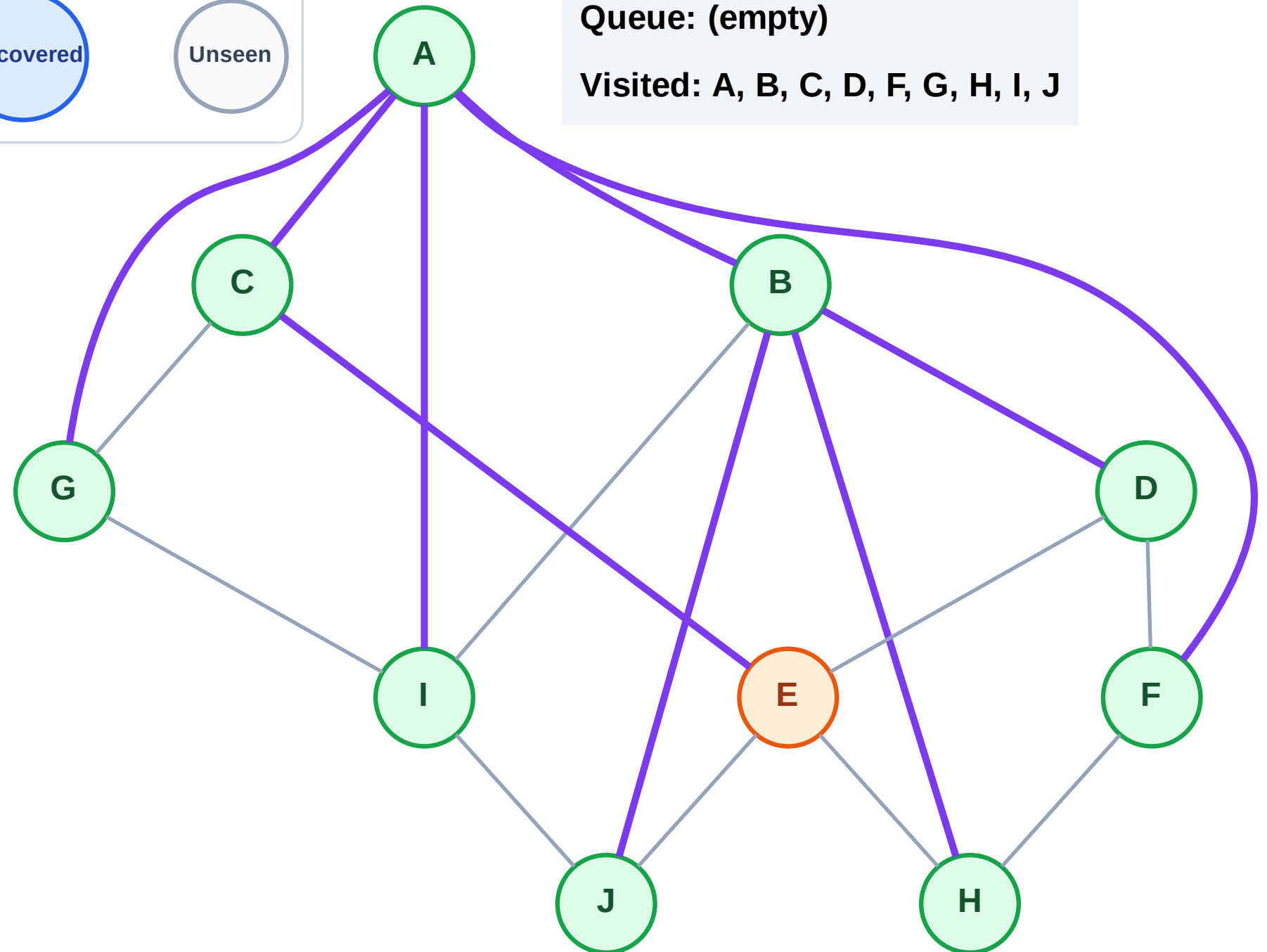
Step 20: Process node E

Legend



Queue: (empty)

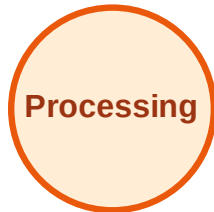
Visited: A, B, C, D, F, G, H, I, J



BFS Traversal

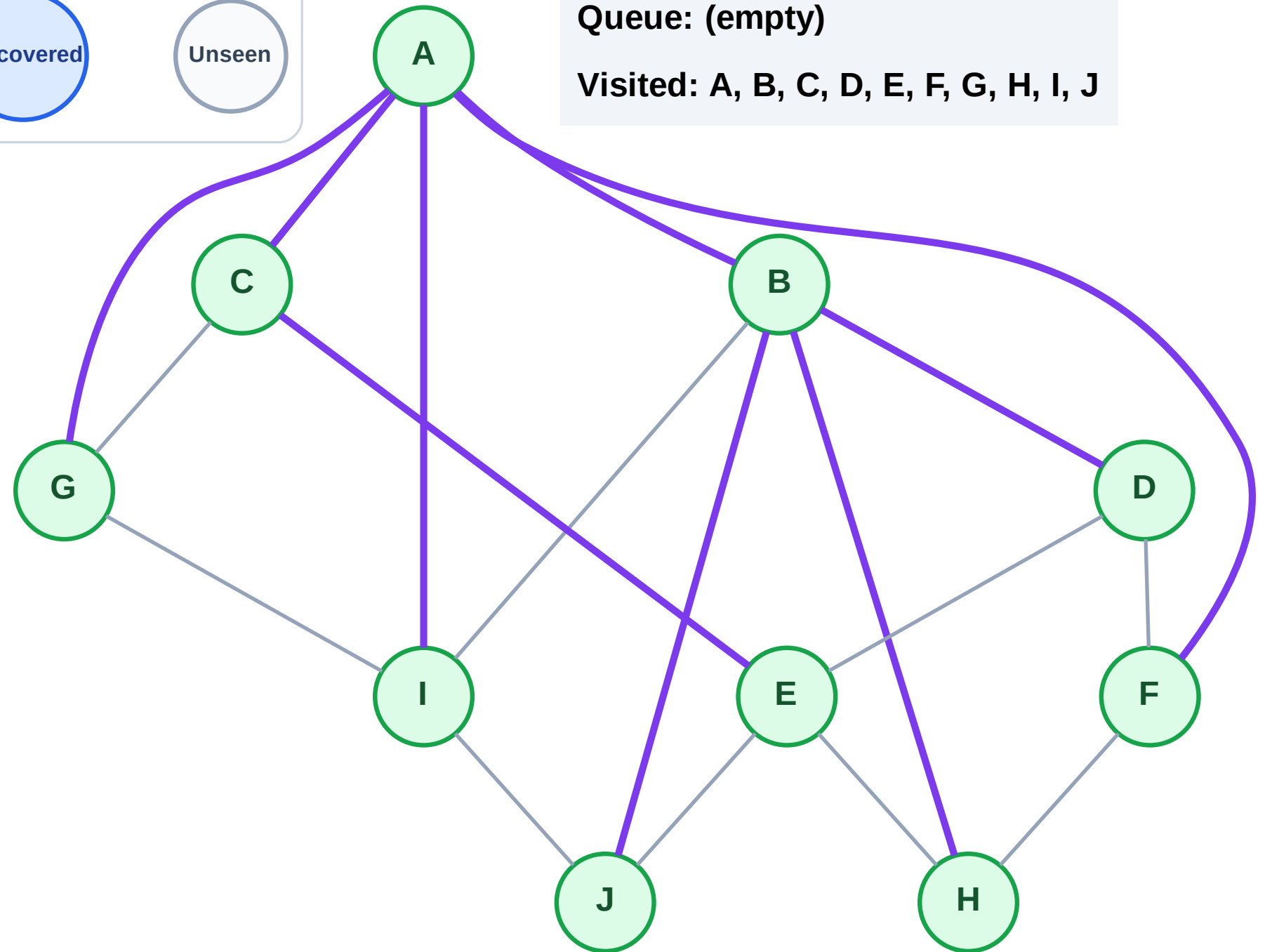
Step 21: No new neighbors from E

Legend



Queue: (empty)

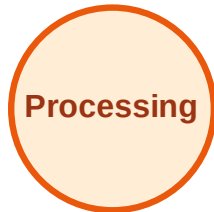
Visited: A, B, C, D, E, F, G, H, I, J



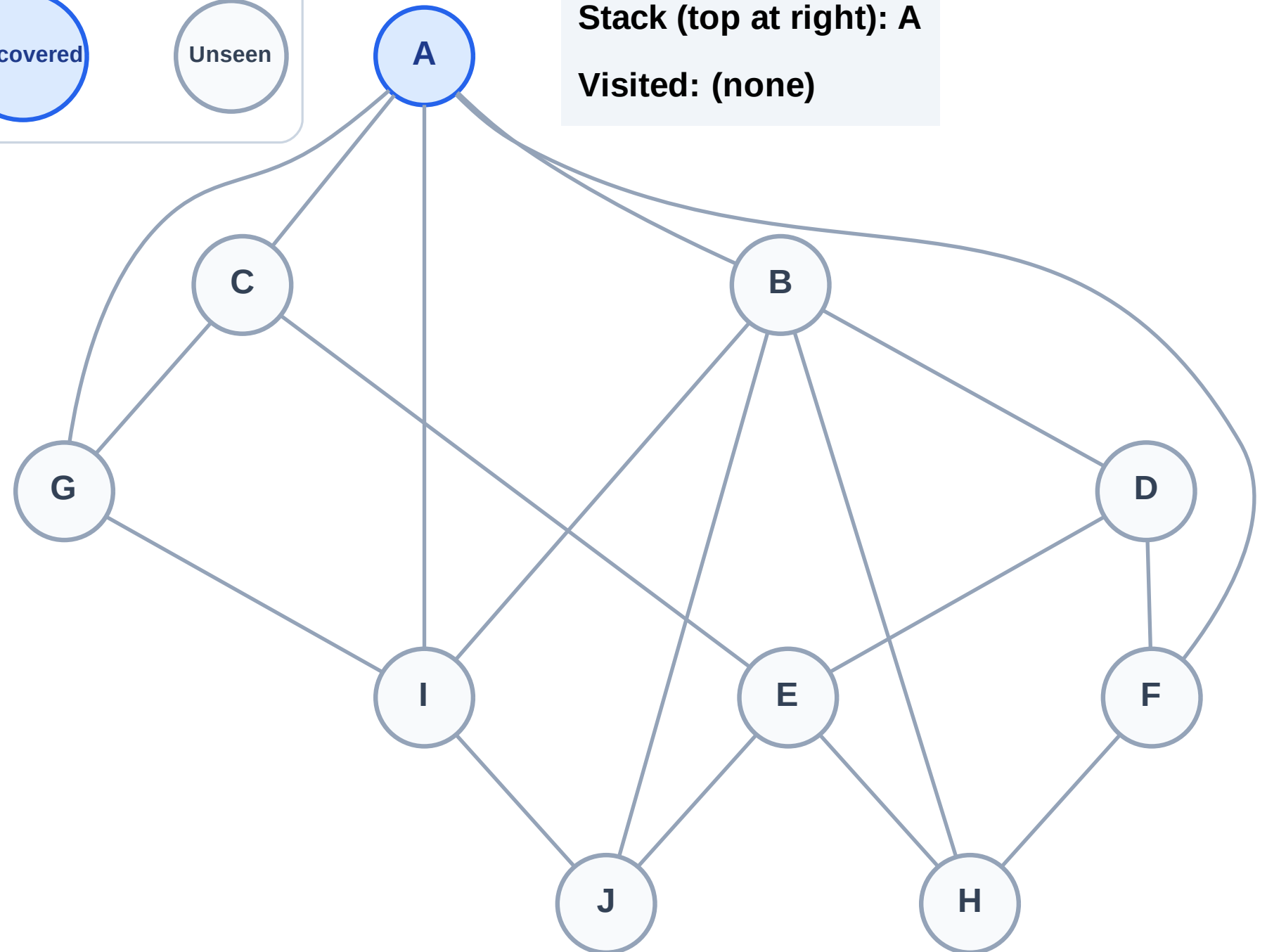
DFS Traversal

Step 1: Start at A

Legend



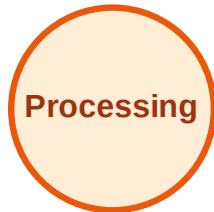
Stack (top at right): A
Visited: (none)



DFS Traversal

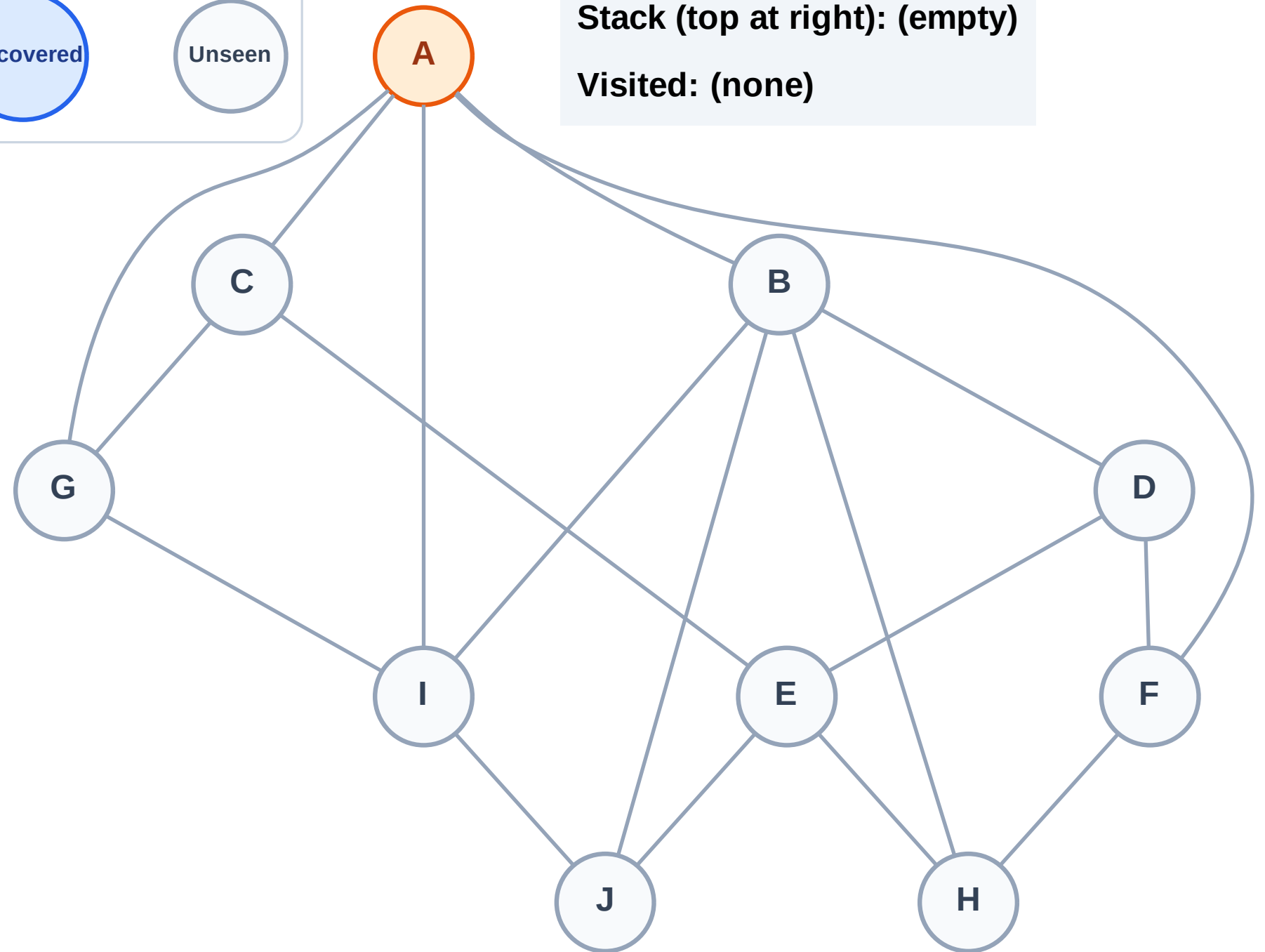
Step 2: Process node A

Legend



Stack (top at right): (empty)

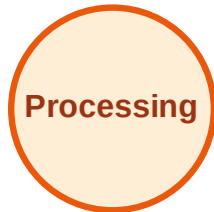
Visited: (none)



DFS Traversal

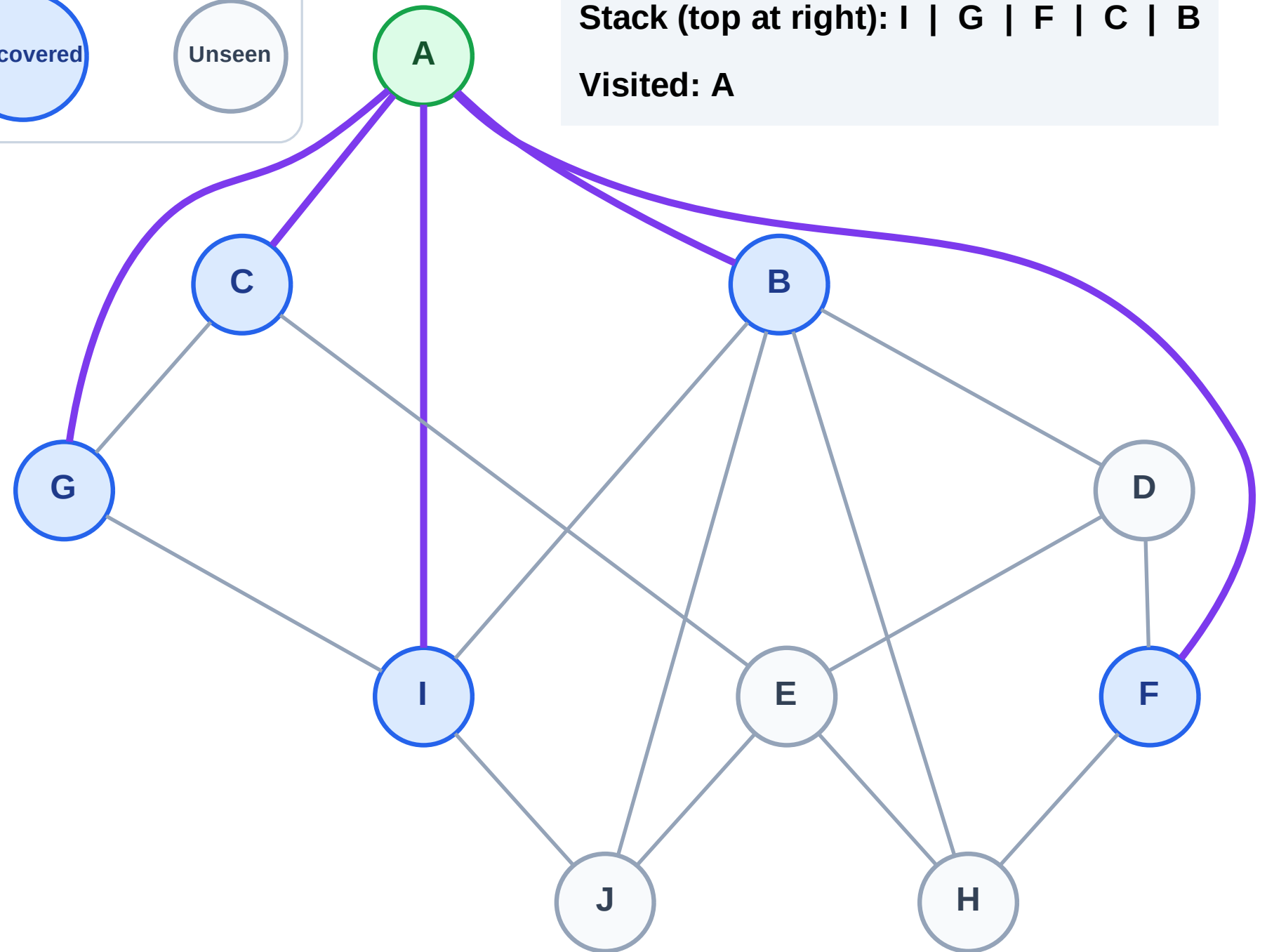
Step 3: Discover I, G, F, C, B from A

Legend



Stack (top at right): I | G | F | C | B

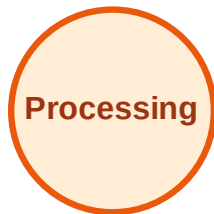
Visited: A



DFS Traversal

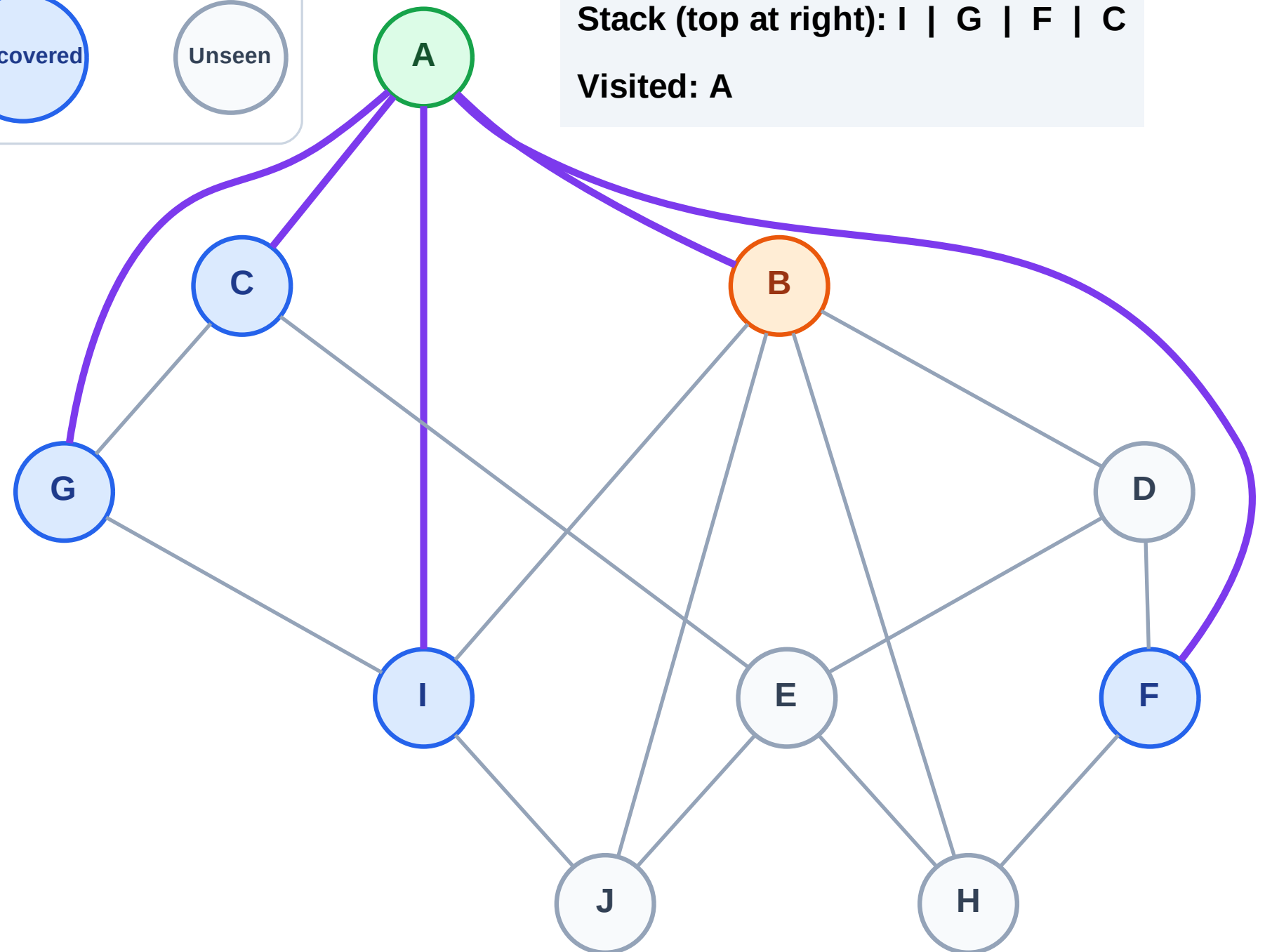
Step 4: Process node B

Legend



Stack (top at right): I | G | F | C

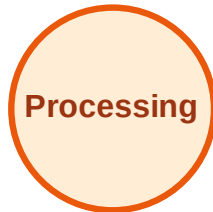
Visited: A



DFS Traversal

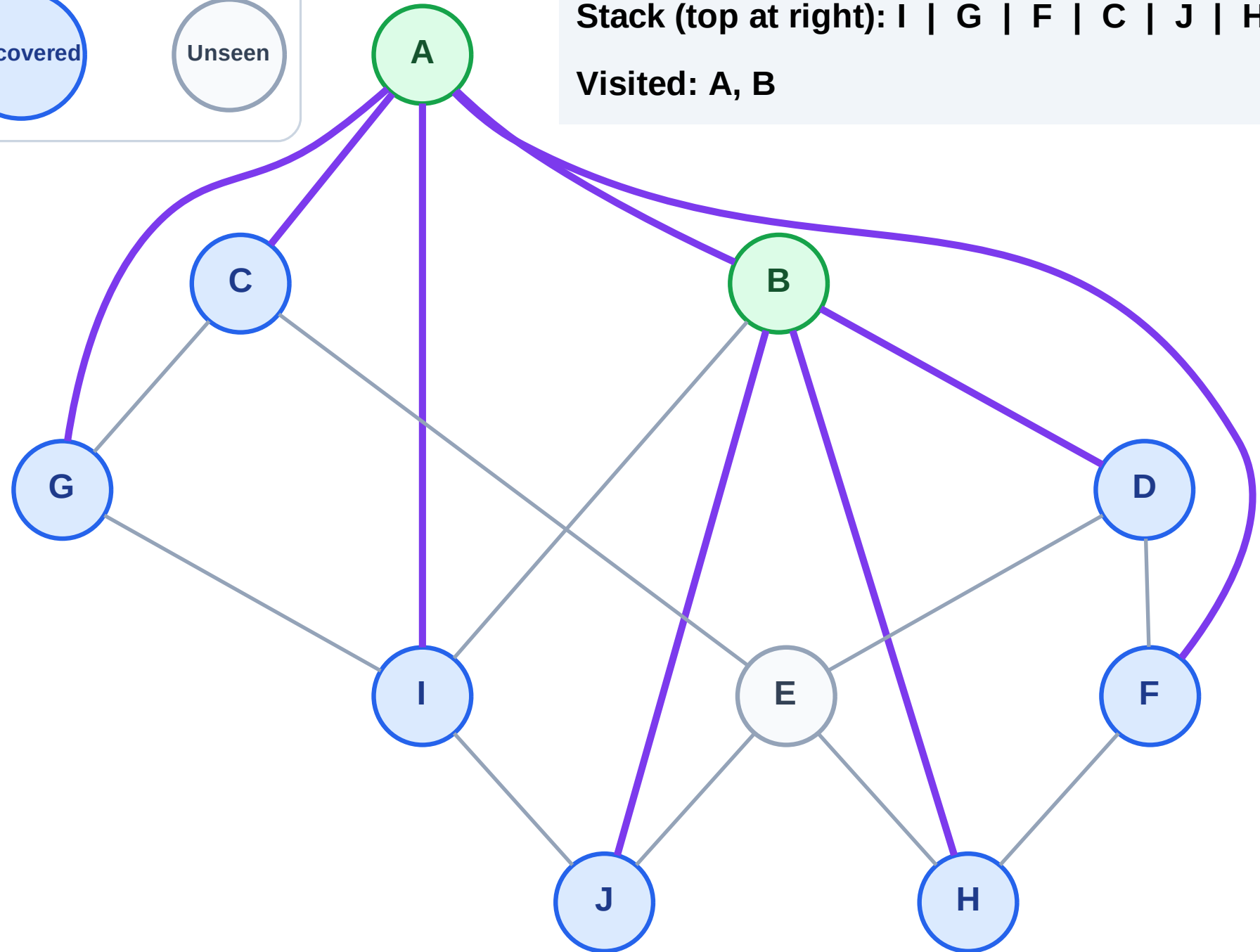
Step 5: Discover J, H, D from B

Legend



Stack (top at right): I | G | F | C | J | H | D

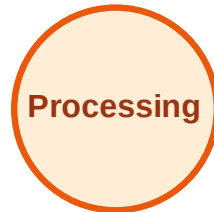
Visited: A, B



DFS Traversal

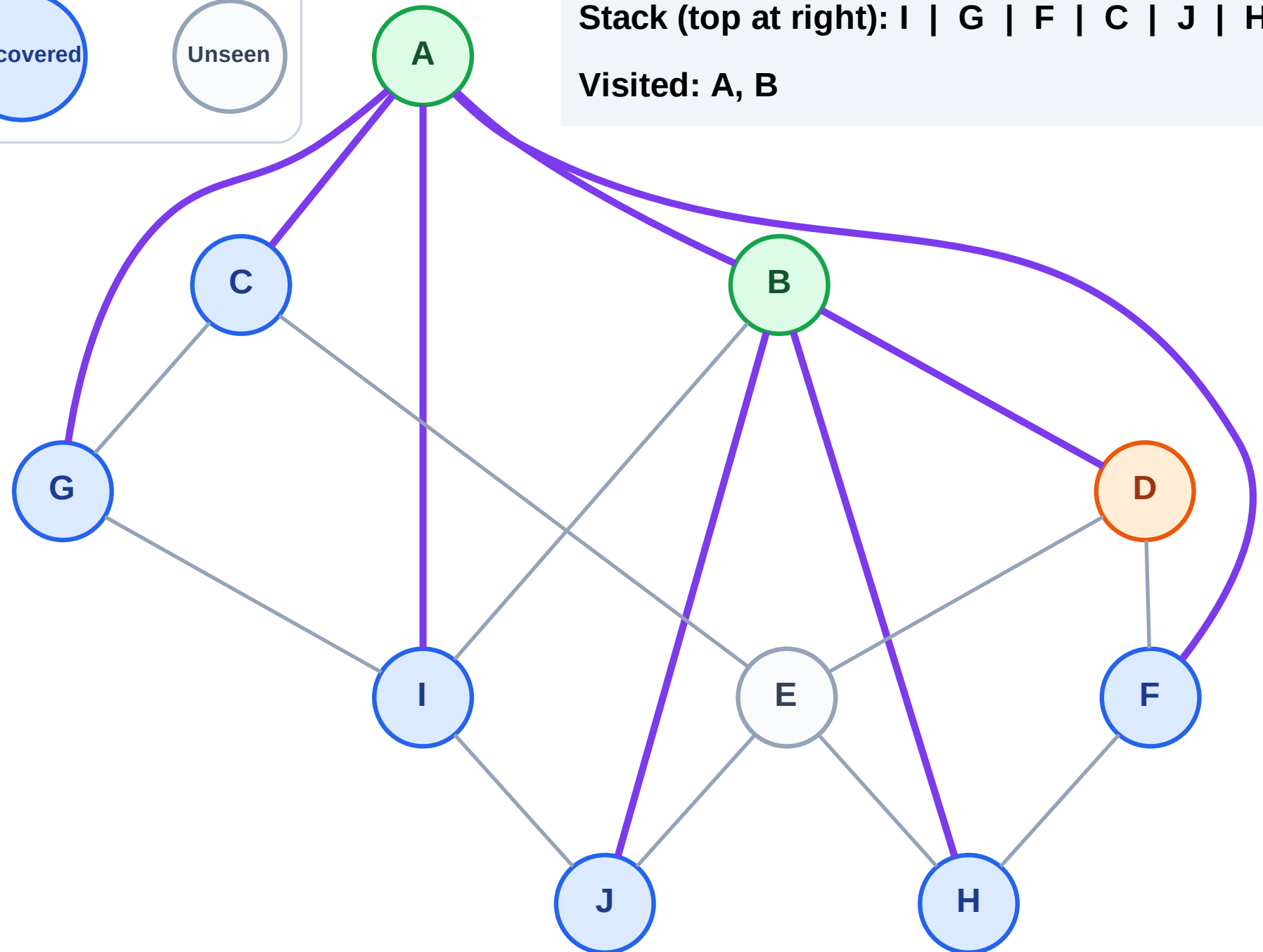
Step 6: Process node D

Legend



Stack (top at right): I | G | F | C | J | H

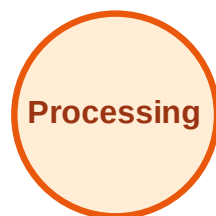
Visited: A, B



DFS Traversal

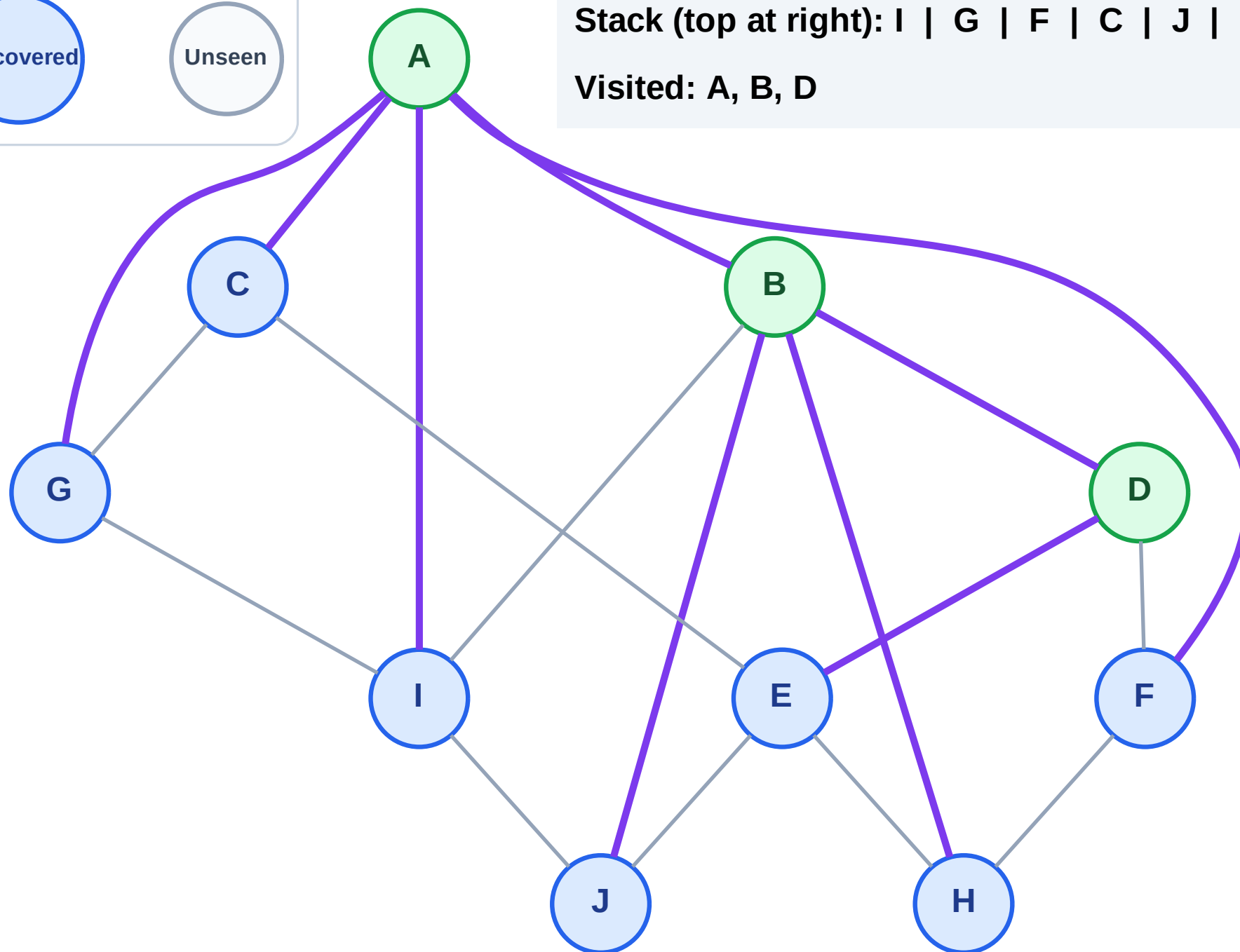
Step 7: Discover E from D

Legend



Stack (top at right): I | G | F | C | J | H | E

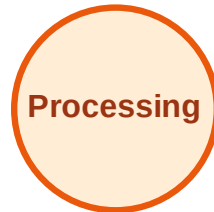
Visited: A, B, D



DFS Traversal

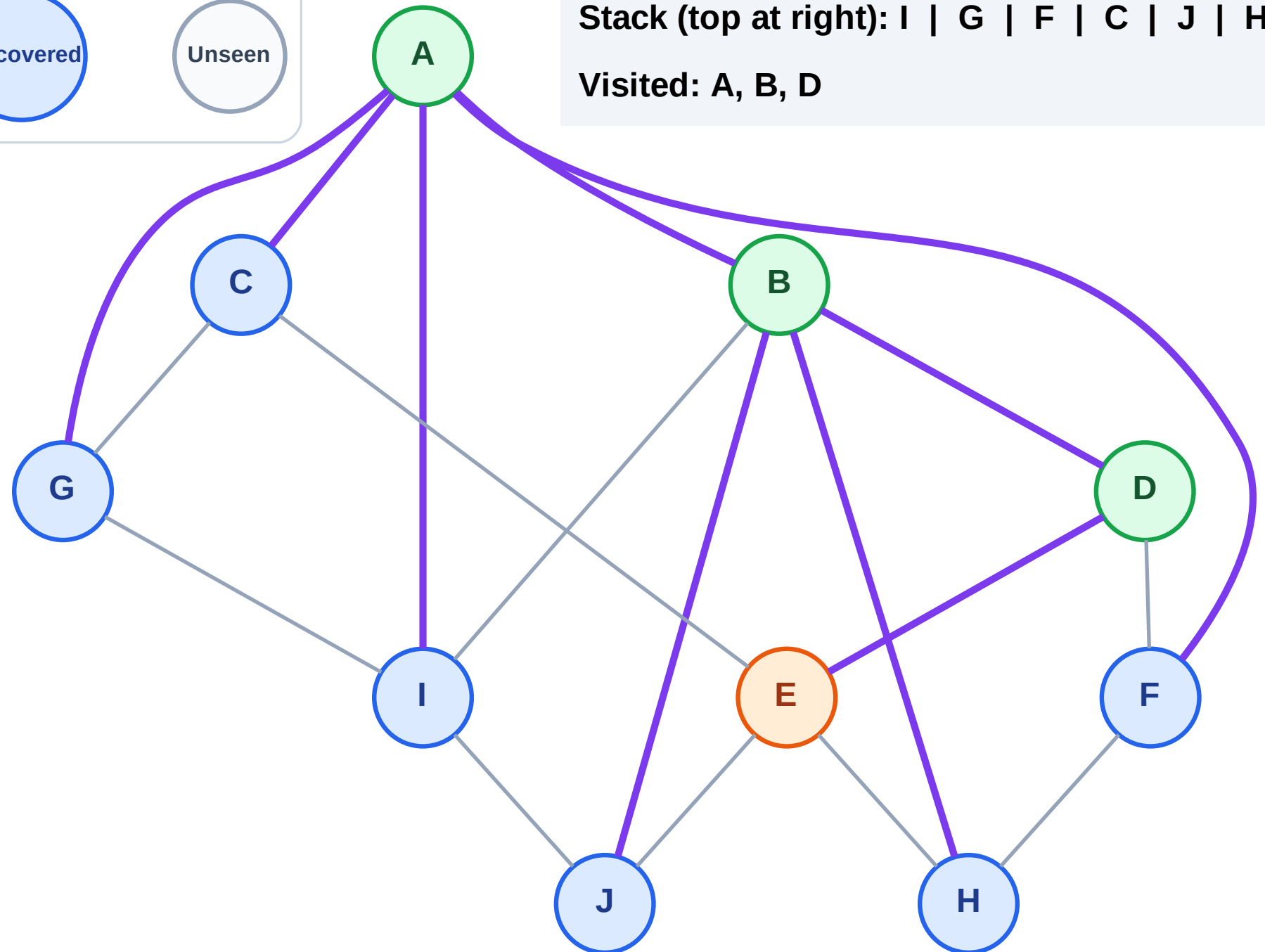
Step 8: Process node E

Legend



Stack (top at right): I | G | F | C | J | H

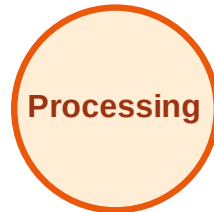
Visited: A, B, D



DFS Traversal

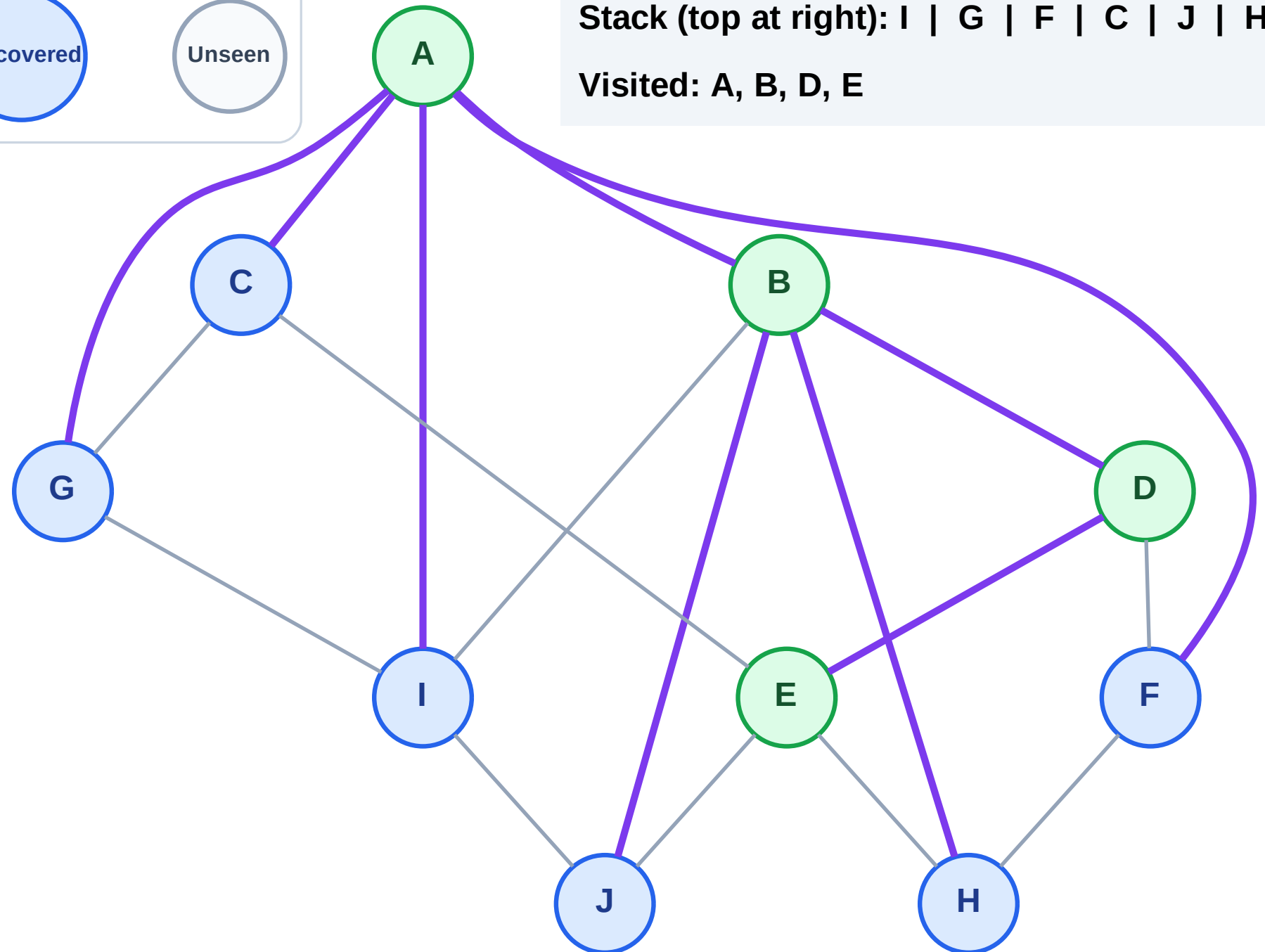
Step 9: No new neighbors from E

Legend



Stack (top at right): I | G | F | C | J | H

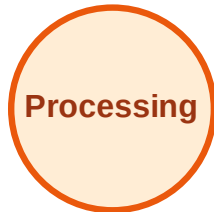
Visited: A, B, D, E



DFS Traversal

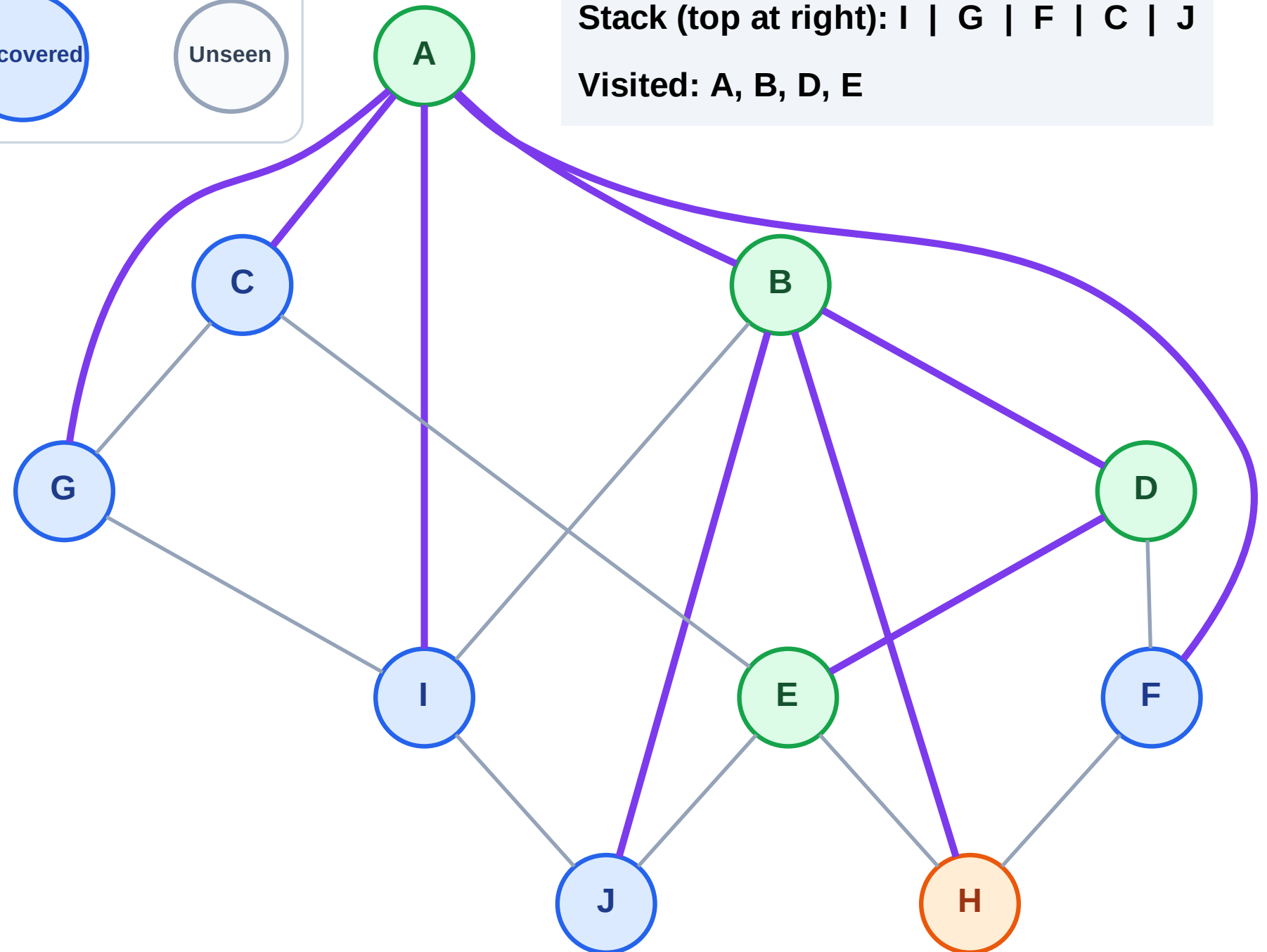
Step 10: Process node H

Legend



Stack (top at right): I | G | F | C | J

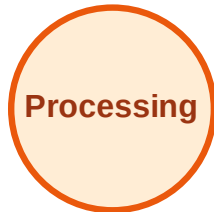
Visited: A, B, D, E



DFS Traversal

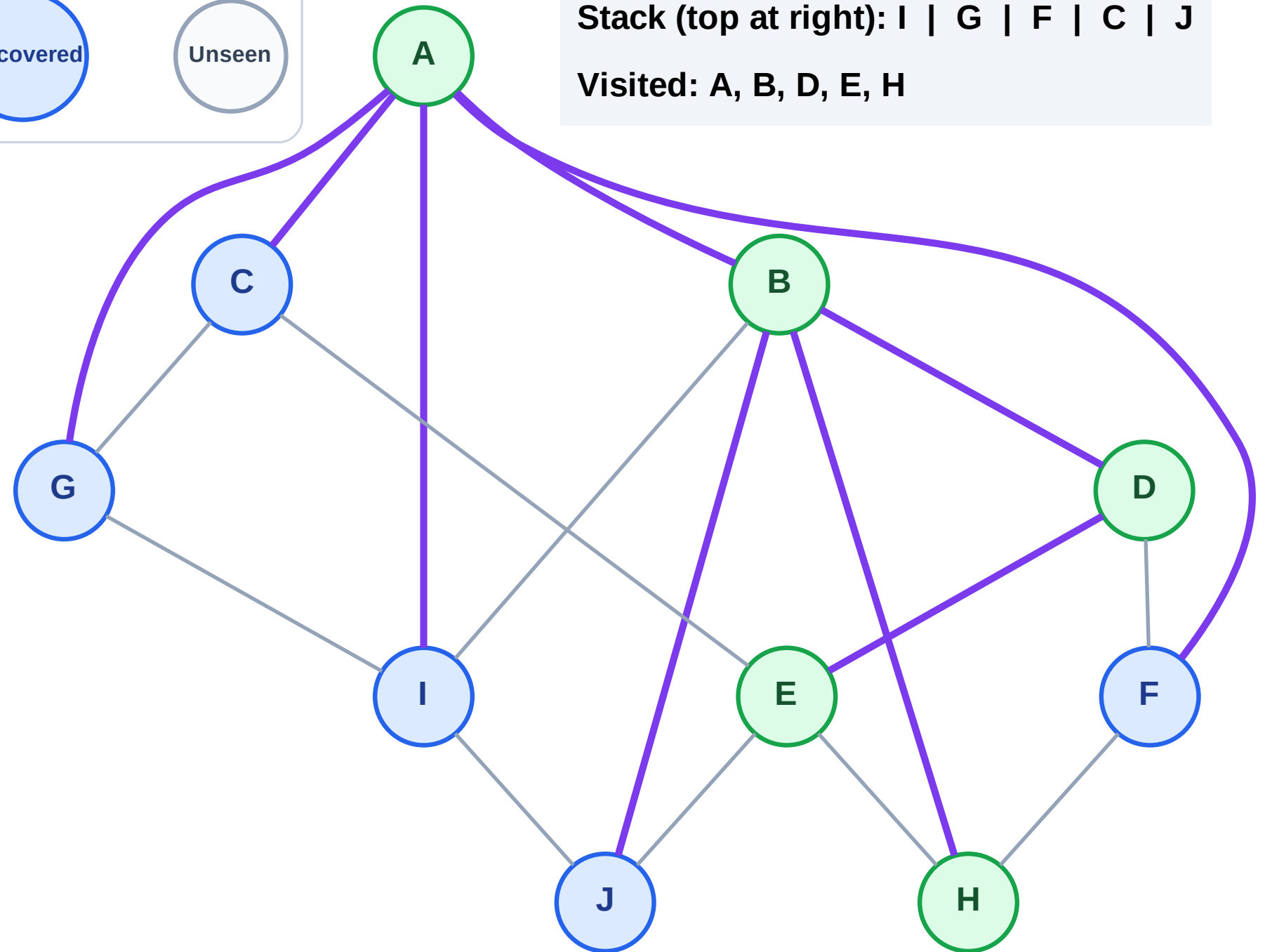
Step 11: No new neighbors from H

Legend



Stack (top at right): I | G | F | C | J

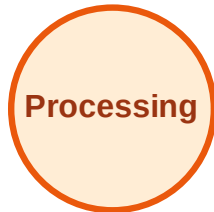
Visited: A, B, D, E, H



DFS Traversal

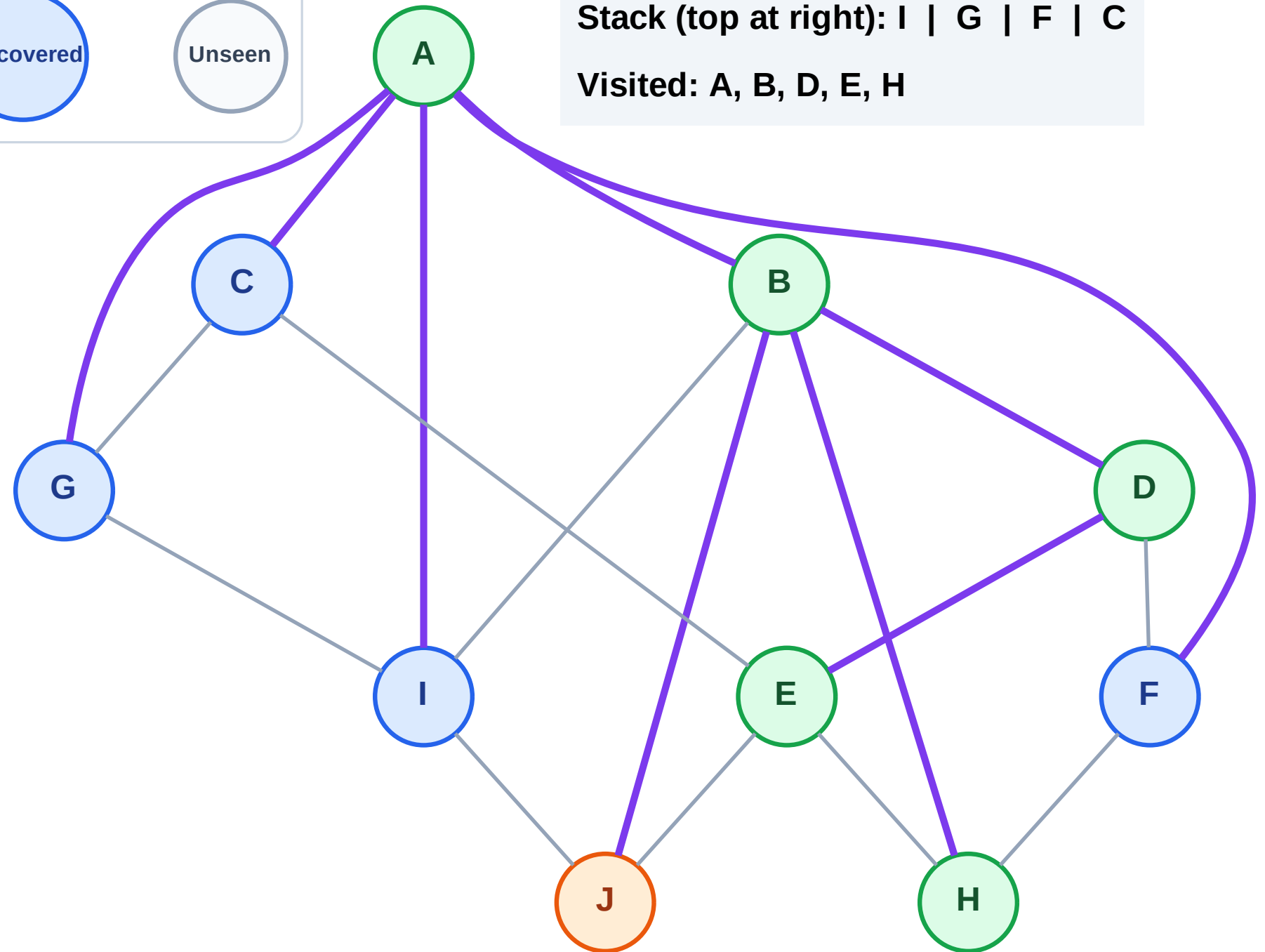
Step 12: Process node J

Legend



Stack (top at right): I | G | F | C

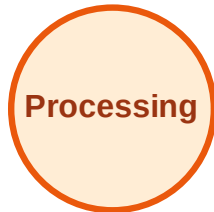
Visited: A, B, D, E, H



DFS Traversal

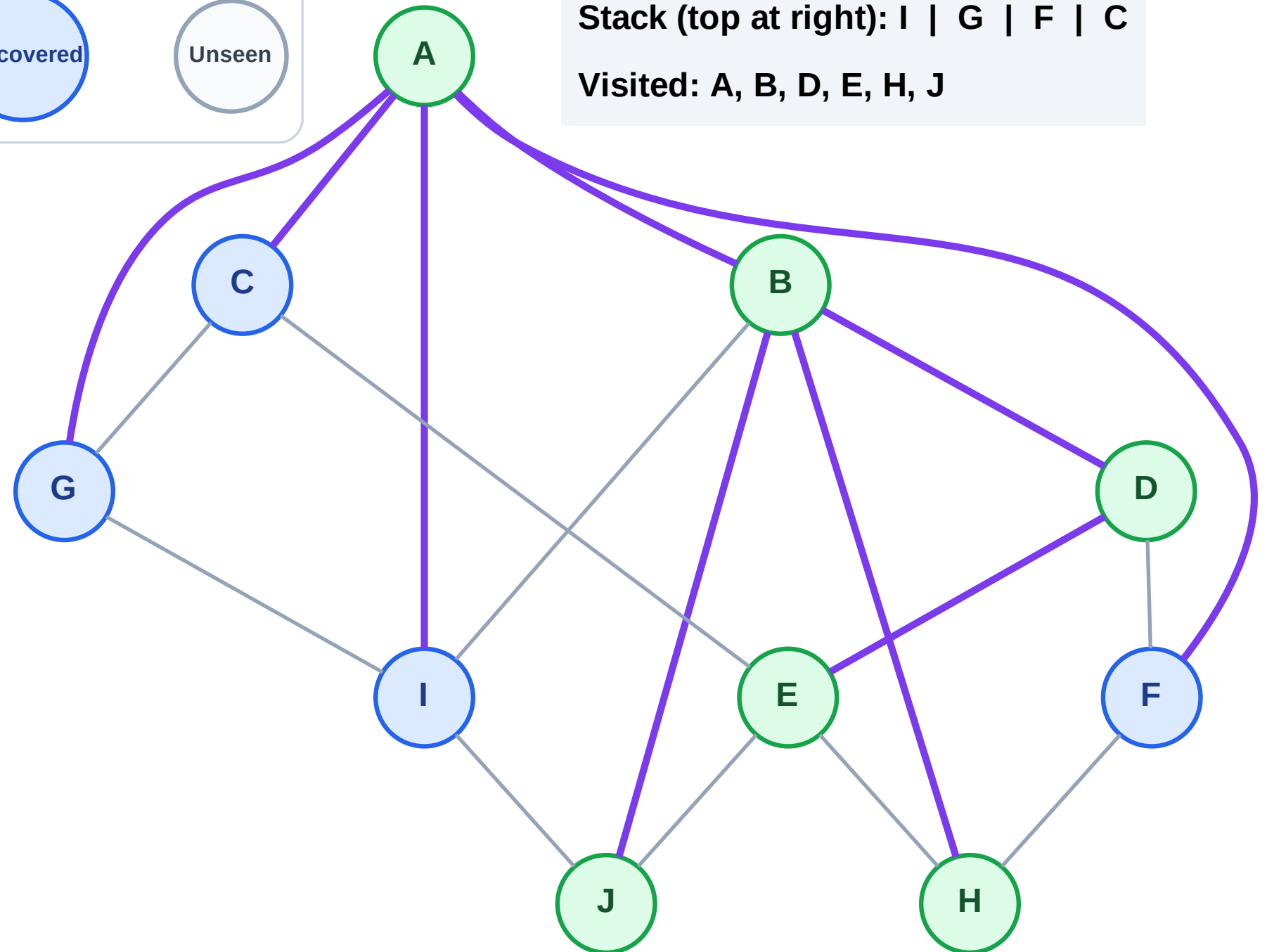
Step 13: No new neighbors from J

Legend

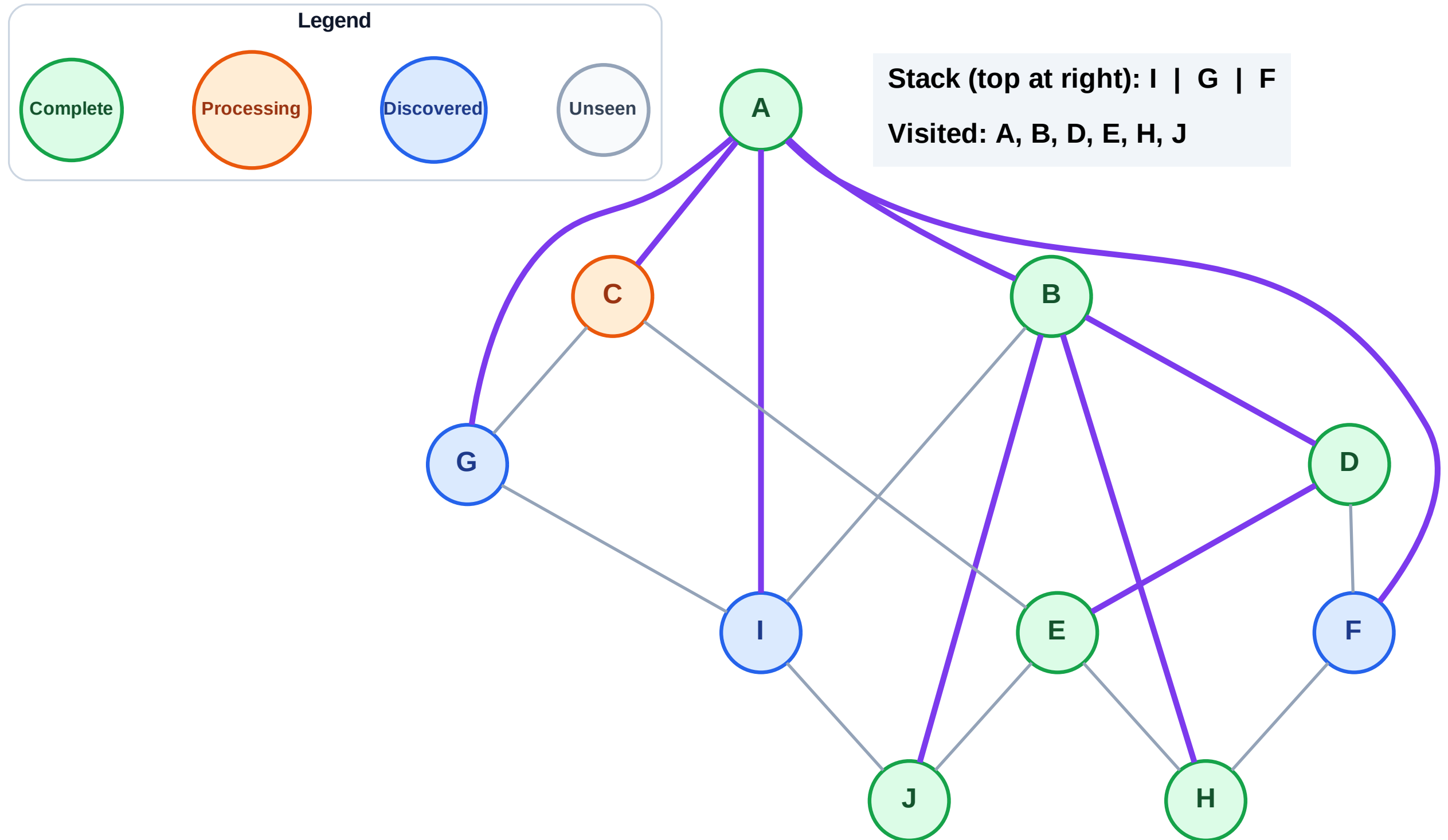


Stack (top at right): I | G | F | C

Visited: A, B, D, E, H, J

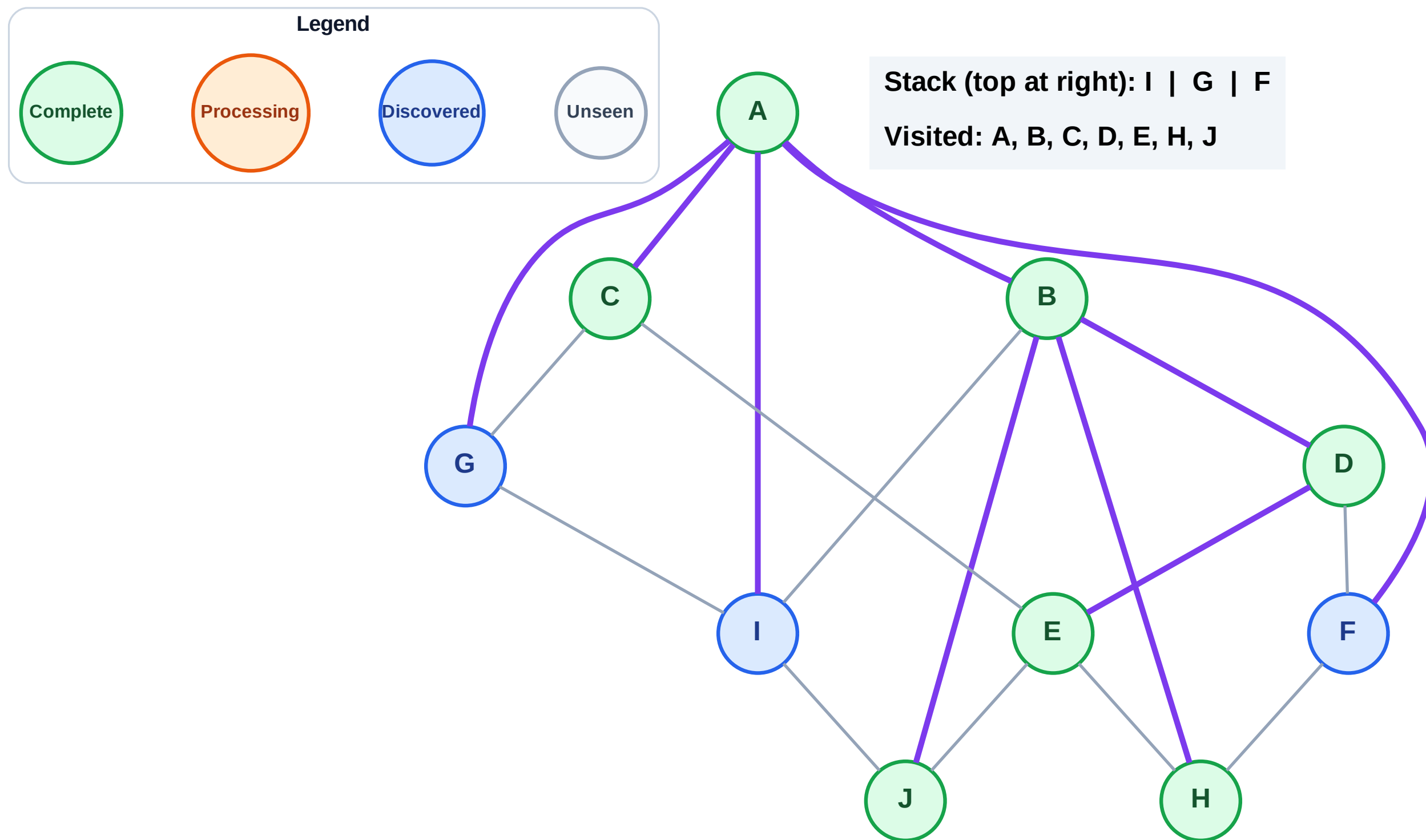


Step 14: Process node C



Step 15: No new neighbors from C

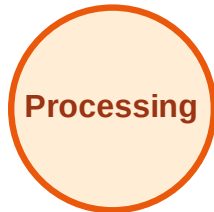
Step 15: No new neighbors from C



DFS Traversal

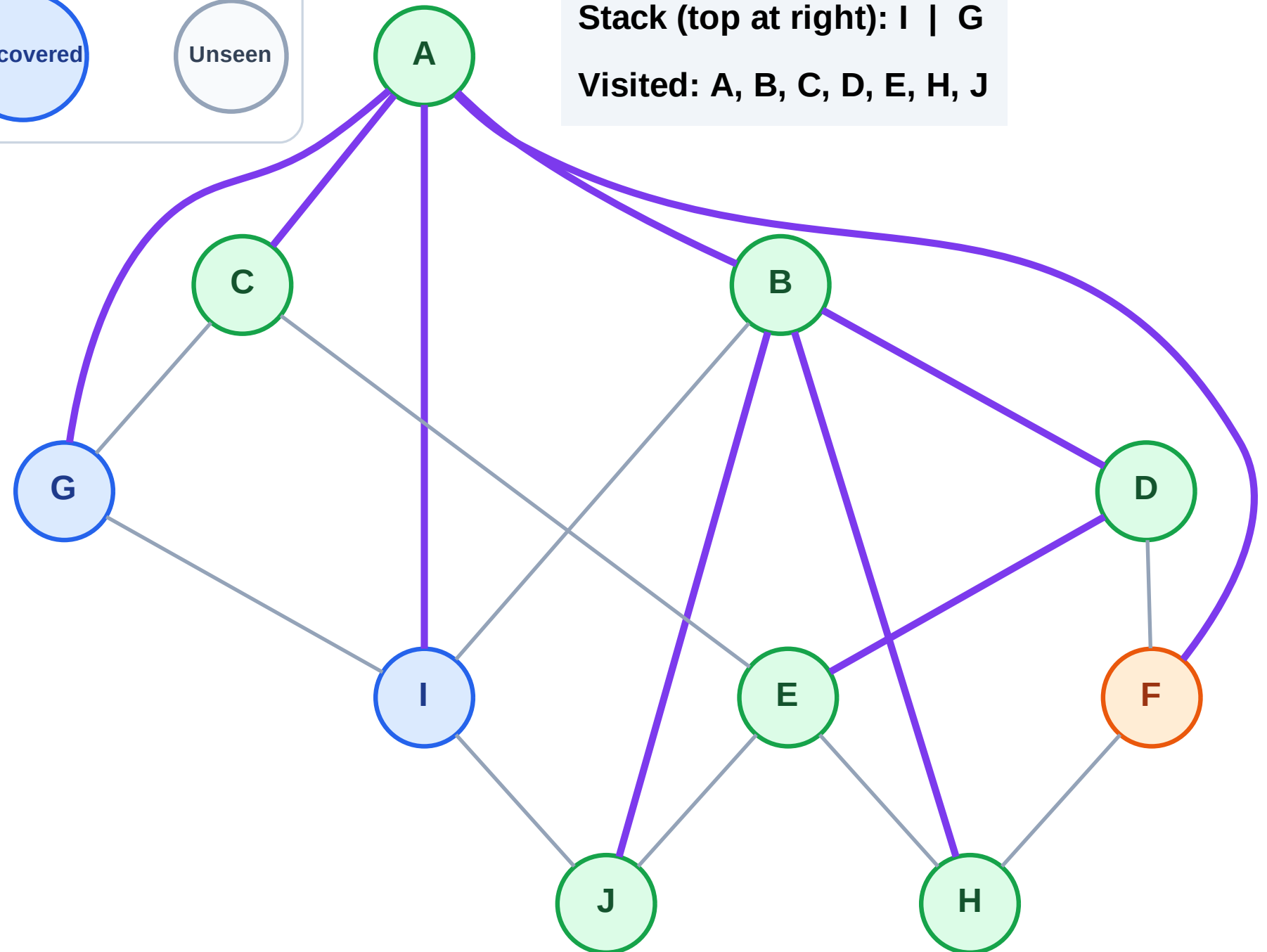
Step 16: Process node F

Legend



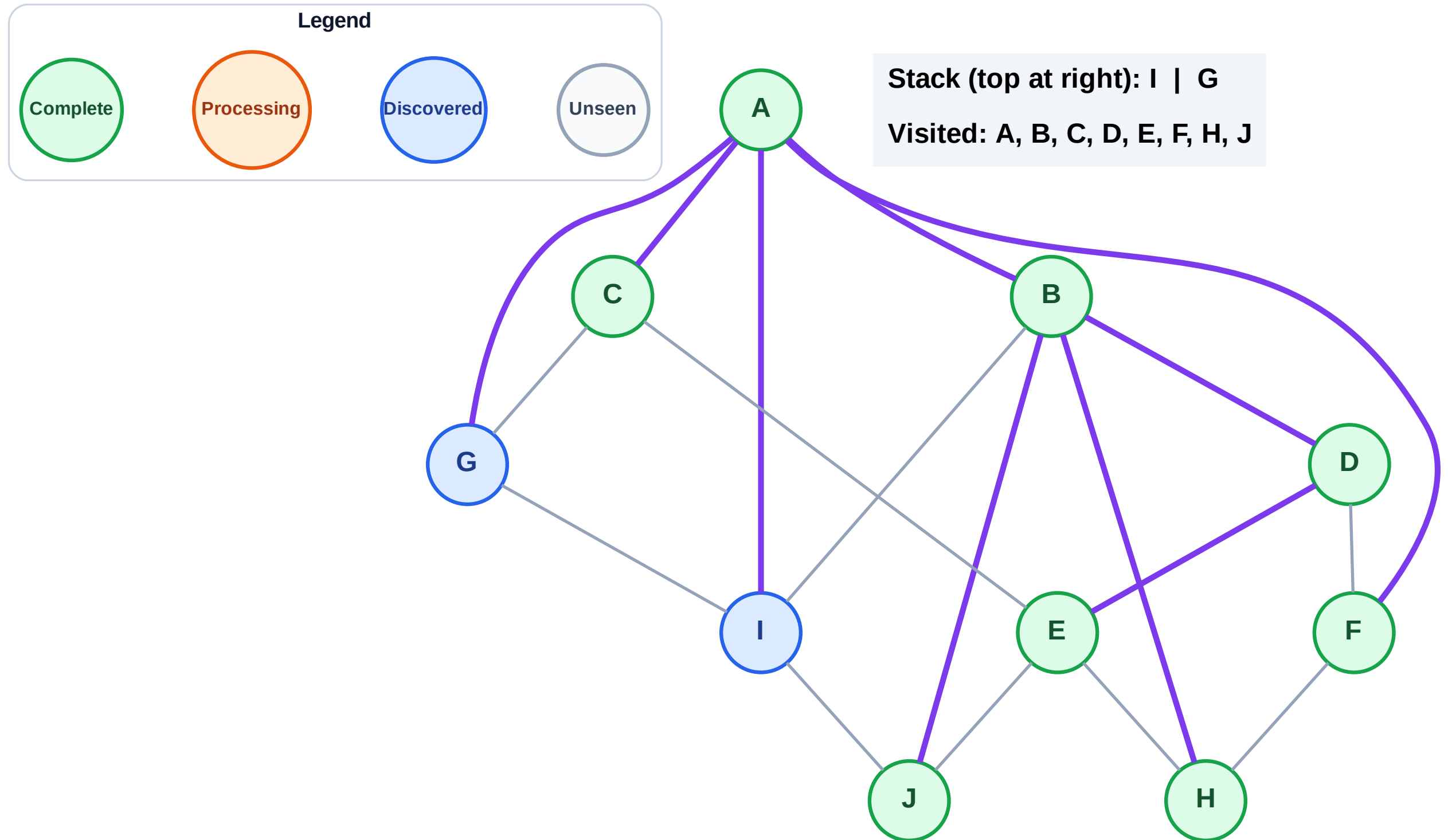
Stack (top at right): I | G

Visited: A, B, C, D, E, H, J



Step 17: No new neighbors from F

Step 17: No new neighbors from F



DFS Traversal

Step 18: Process node G

Legend

Complete

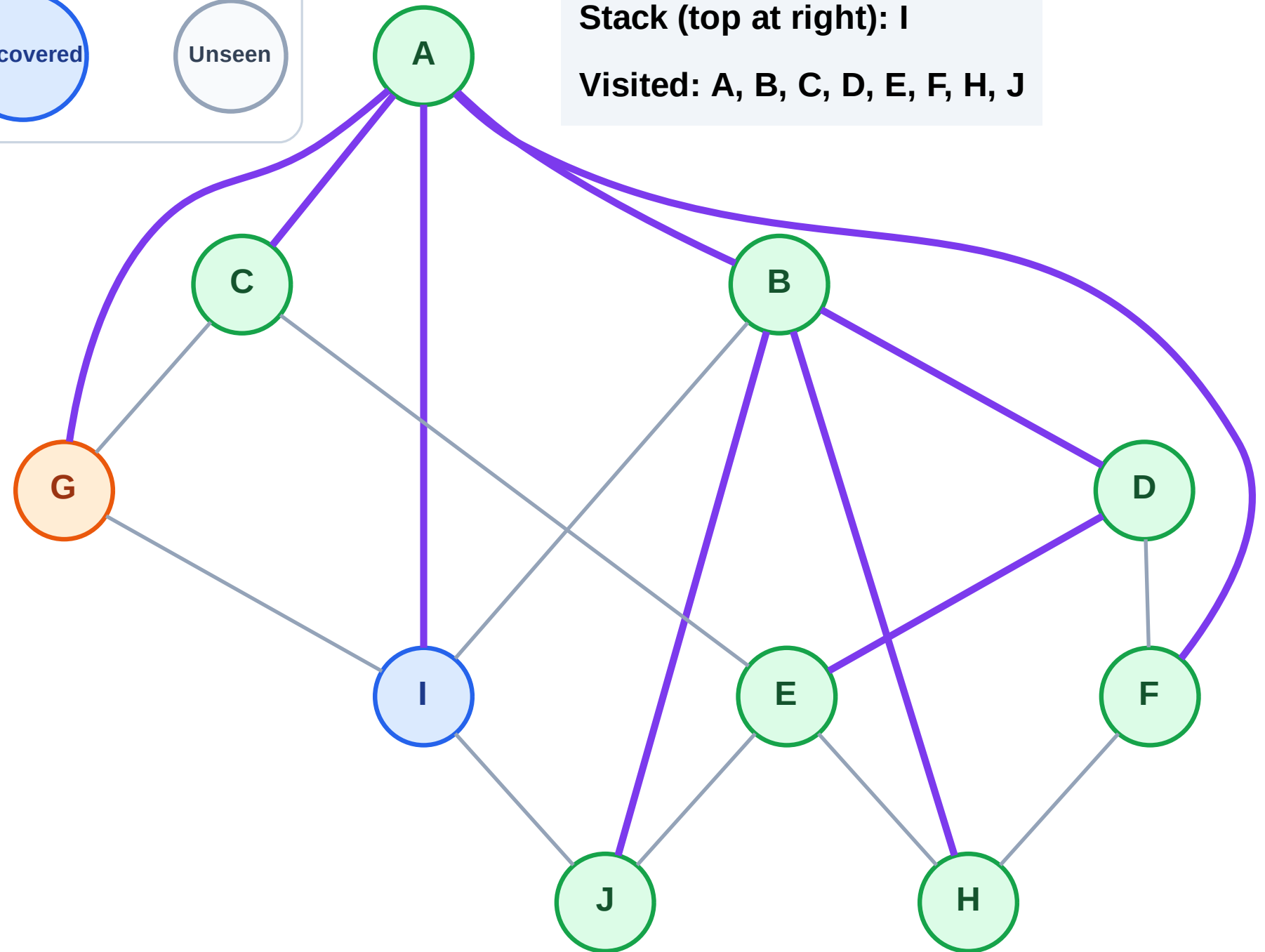
Processing

Discovered

Unseen

Stack (top at right): I

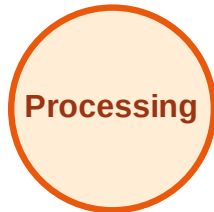
Visited: A, B, C, D, E, F, H, J



DFS Traversal

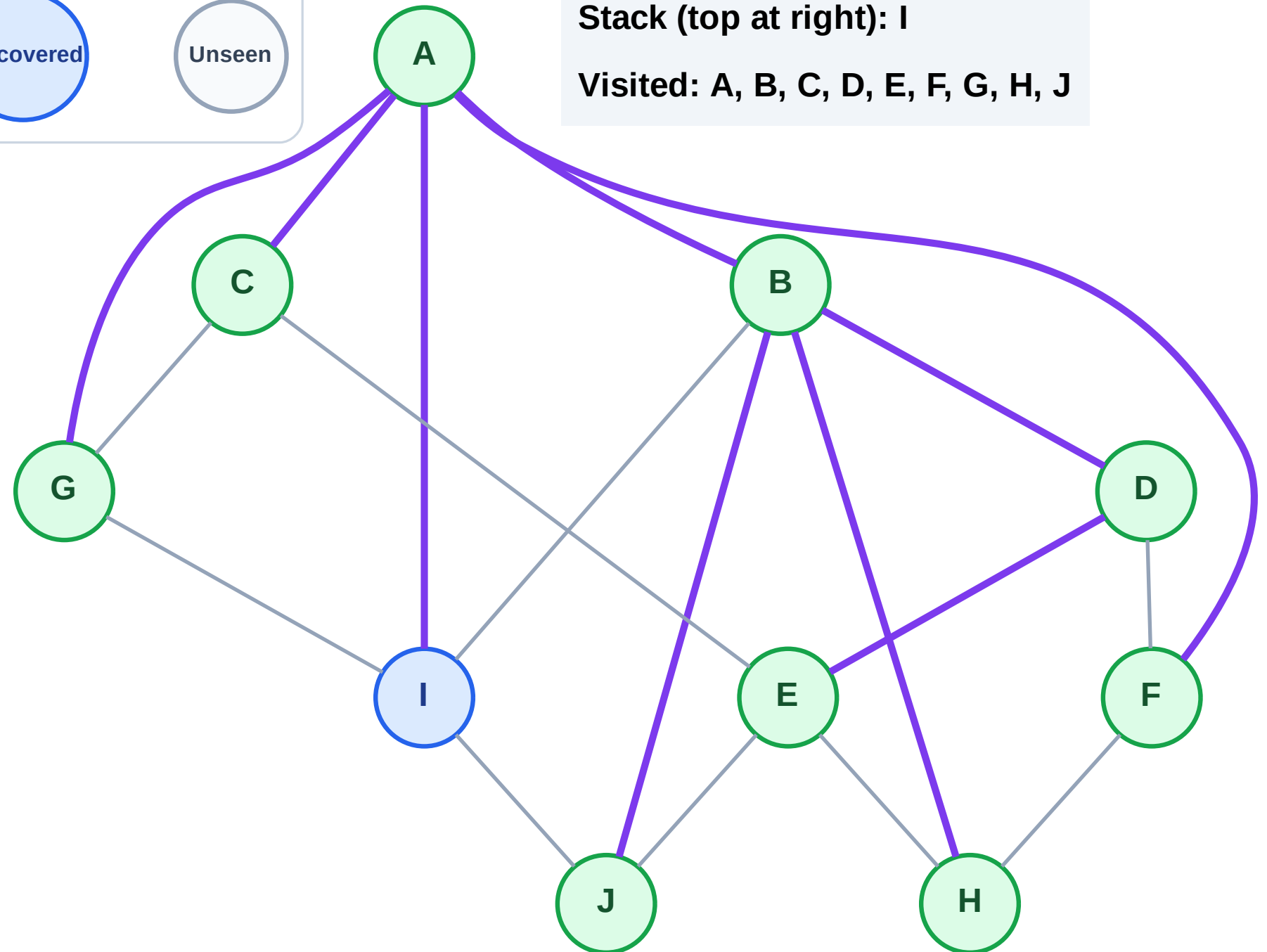
Step 19: No new neighbors from G

Legend



Stack (top at right): I

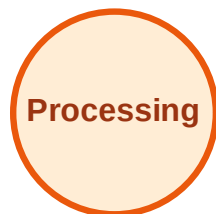
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

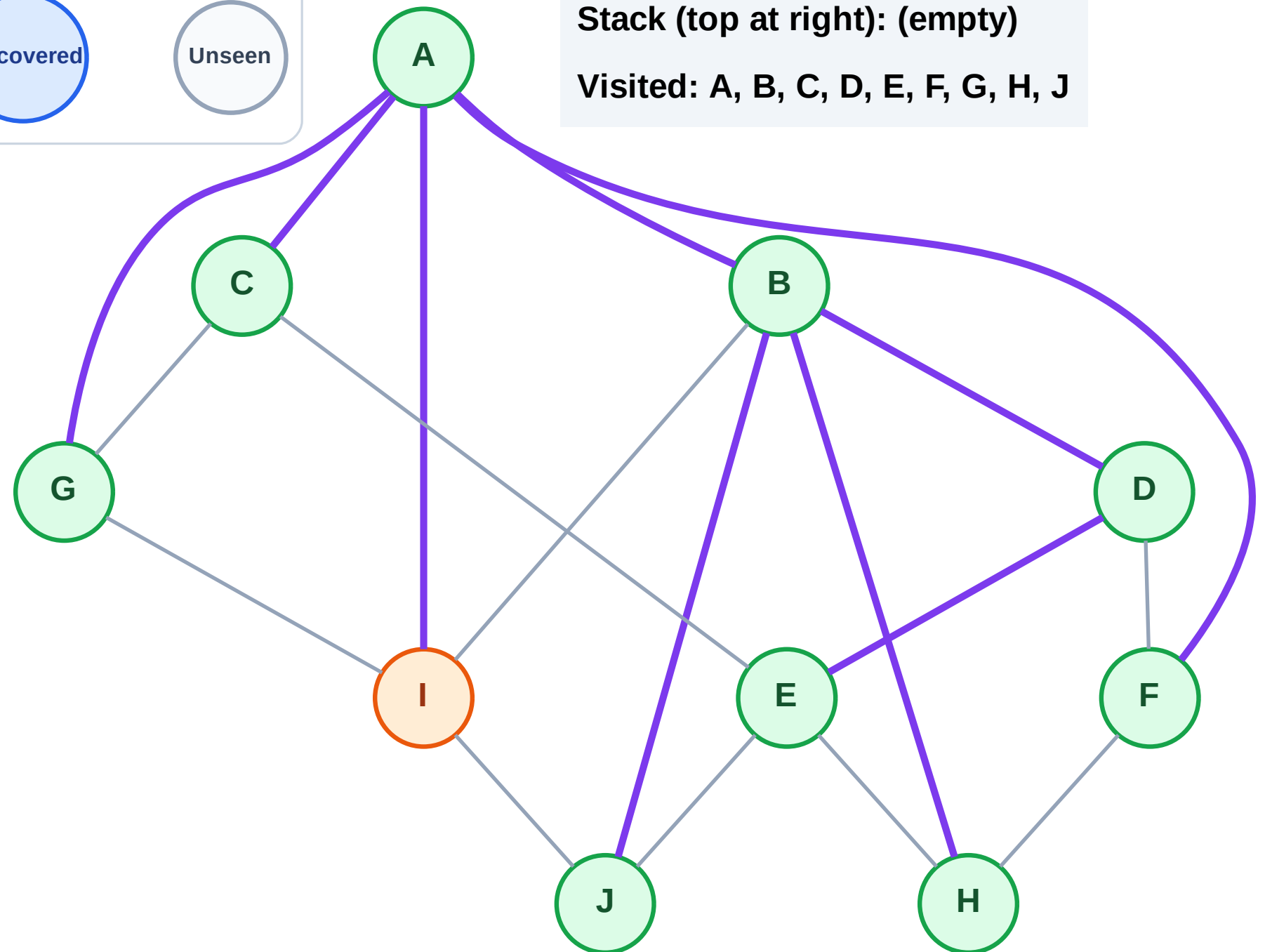
Step 20: Process node I

Legend



Stack (top at right): (empty)

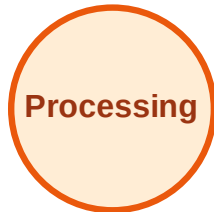
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

